Introduction to Work Measurement

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Time is important in work systems because of its economic significance. Most workers are paid for the time they are on the job. The labor content to produce a product or deliver a service is often a major determinant of the cost of the product or service. For any organization to operate efficiently and effectively, it is critical to know how much time should be required to accomplish a given amount of work. For the reader's amusement, a list of quotations and sayings about time is compiled in Table 1.

The terms *time study* and *work measurement* are often used interchangeably. Both are concerned with how much time it should take to complete a unit of work. ***Work measurement*** refers to a set of four techniques that are concerned with the evaluation of a task in terms of the time that should be allowed for an average human worker to perform that task: (1) direct time study, (2) predetermined motion time systems, (3) standard data systems, and (4) work sampling, an alternative work measurement technique in which statistical measures are determined about how workers allocate their time among

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TABLE 1 Quotations and Sayings About Time

Time flies when you're having fun.
Take care of the minutes, and the hours will take care of themselves. (Lord Chesterfield)
Time heals all things—except leaky faucets.
Those who kill time eventually mourn the corpse.
Time changes with age: in youth, time marches on; in middle age, time flies; and in old age, time runs out.
All things come to him who waits, except the precious time lost while waiting.
Some people count time, others make time count.

Source: Compiled from E. Esar, *20,000 Quips and Quotes* (New York: Barnes and Noble 1995) and other sources.

multiple activities. The task that is measured is usually a repetitive work cycle. The objective of these work measurement techniques is to determine a ***standard time*** for the task.

Because of its emphasis on time, work measurement is often called time study. However, in modern usage and in our coverage in this text, time study has a slightly broader meaning. Whereas work measurement is generally focused on human work effort, ***time study*** refers to all of the ways in which time is investigated and analyzed in work situations, whether the work is accomplished by human workers or automated systems. For example, in an automated work system in which a human worker periodically attends to the system but is not present every cycle, it is still important to know the automatic cycle time. Time study also includes the learning curve phenomenon, which is not usually included in the definition of work measurement.

Here, we discuss the various techniques used to measure work and other job activities of human workers. Time is an important variable used to evaluate performance in the variety of work systems. In manual work, worker-machine systems, assembly lines, logistics work, service operations, and projects, time is the common factor that determines performance. In this chapter, we provide an overview of work measurement, the basic definitions and principles that apply to all of the techniques, and the issues that are faced in their application.

1 TIME STANDARDS AND HOW THEY ARE DETERMINED

Most workers are paid on the basis of time. The common work shift is 8 hours per day, and the worker is paid an hourly rate for those 8 hours. Several questions may be asked by a worker's employer: How much work did the worker accomplish in those 8 hours? Has the worker done a fair day's work? Time standards provide a way to answer these questions.[1] The most meaningful and useful measure of work is the amount of time it takes to accomplish it. Time is objective. It is quantifiable. People understand time.

[1]The terms *time standard* and *standard time* are used interchangeably and synonymously in this text. Other terms for time standards include *work standards* and *labor standards*.

The ***standard time*** for a given task is the amount of time that should be allowed for an average worker to process one work unit using the standard method and working at a normal pace. The standard time includes some additional time, called the ***allowance***, to provide for the worker's personal needs, fatigue, and unavoidable delays during the shift. The standard time is sometimes referred to as the ***allowed time***, because it indicates how much time is allowed the worker to process each work unit so that by the end of the shift a fair day's workload has been accomplished, despite the various interruptions that may occur.

How does an organization know whether it needs time standards for its operations? The following characteristics are typical of industrial situations in which time standards would be beneficial:

- *Low productivity.* If the current level of productivity is low, then there are significant opportunities for improvement.
- *Repeat orders.* Once the time standard is set during the first order, the same standard can be used for successive orders.
- *Long production runs.* A long production run means that the time invested to set the standard is apportioned over more parts, thus reducing the average cost of work measurement.
- *Repetitive work cycles.* Work measurement can be justified more readily when the work cycle is highly repetitive.
- *Short cycle times.* Short work cycles require less time to set standards.

When accurately determined, time standards serve the following important functions and applications in an organization:

- They define a "fair day's work." If the quantity of work units completed by a worker, when multiplied by their respective standard times, adds up to the number of hours in the shift, then the worker has put in a fair day's work.
- They provide a means of converting the workload to be produced into the staffing and equipment resources needed to accomplish the workload. They help to determine manpower requirements and capacity limitations.
- They provide an objective way to compare alternative methods for accomplishing the same task.
- They provide a basis for wage incentives and for evaluating worker performance.
- They provide time data for production planning and scheduling, cost estimating, material requirements planning, and other management functions that depend on accurate task time data.

1.1 Methods to Determine Time Standards

There are several methods by which "standard times" can be established for a task. They vary in terms of the accuracy and reliability of the values that are derived from them

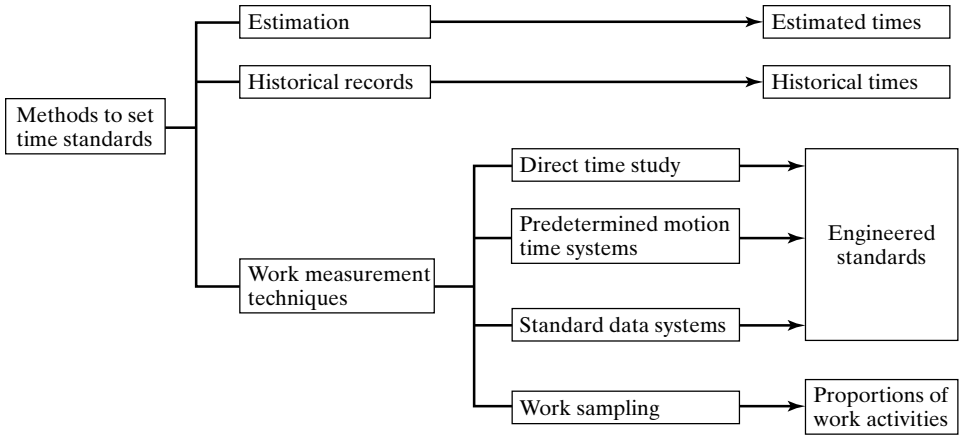


Figure 1 Classification of methods to determine time standards.

and in the amount of time required to apply them. The following methods can be used to determine time standards, as shown in Figure 1:

- *Estimation.* In this method, the department foreman or other person familiar with the jobs performed in the department is asked to judge how much time should be allowed for the given task. Because this method depends on the estimator's judgment, it is the least accurate of the techniques for determining time standards.[2] On the other hand, it is better than making no estimate of the time and simply assigning a worker to the task with no accountability.
- *Historical records of previous production runs.* In this method, the actual times and production quantities from records of previous identical or similar job orders are used to determine the time standards. Most companies require workers to submit time records (e.g., "time cards") indicating the time spent on orders or jobs and the quantities of parts turned in against that time. From these records a calculation is made to determine the average time per part, and this can be used as a "time standard" for the next order. Historical records are an improvement over estimates because they represent actual times for amounts of work completed. Their limitation is that they do not include any indication of the efficiency with which the work was accomplished.
- *Work measurement techniques.* These are the four techniques identified in the chapter introduction: direct time study, predetermined motion time systems, standard data systems, and work sampling. These techniques are described in Section 1.2.

The work measurement techniques are more time consuming to implement, but they are more accurate than estimation or historical records. It is the author's opinion that task time values determined by estimation or historical records should not even be called "time standards." They should be called "estimated times" and "historical times," respectively, and in fact they are sometimes identified by these names.

[2]The issue of accuracy in work measurement is discussed in Section 4.1.

Among the four work measurement techniques, work sampling should be differentiated from the other three. The primary purpose of work sampling is to determine proportions of time spent in various categories of work activity using randomized observations of the subjects of interest. On the other hand, the principal purpose of direct time study, predetermined motion time systems, and standard data systems is to establish time standards. Standards set by these three techniques are sometimes referred to as ***engineered standards***, because (1) they are based on measured time values that have been adjusted for worker performance, and (2) some effort has been made to determine the best method to accomplish the task. Although work sampling can be used to determine time standards, the technique is not as reliable because of statistical errors, and no attempt is made to improve the method by which the work is accomplished.

1.2 Work Measurement Techniques

The four work measurement techniques are described briefly in the following sections. In the pyramidal structure of work, a task consists of work elements, and how each work element consists of basic motion elements. The various work measurement techniques measure work at different levels of this task hierarchy. Figure 2 shows the correspondence between the task hierarchy and work measurement techniques.

Direct Time Study. Direct time study (DTS) involves direct observation of a task using a stopwatch or other chronometric device to record the time taken to accomplish the task. The task is usually divided into work elements and each work element is timed separately. While observing the worker, the time study analyst evaluates the worker's performance (pace), and a record of this pace is attached to each work element time. This evaluation of the worker's pace is called ***performance rating***. The observed time is multiplied by the performance rating to obtain the ***normal time*** for the element or the task:[3]

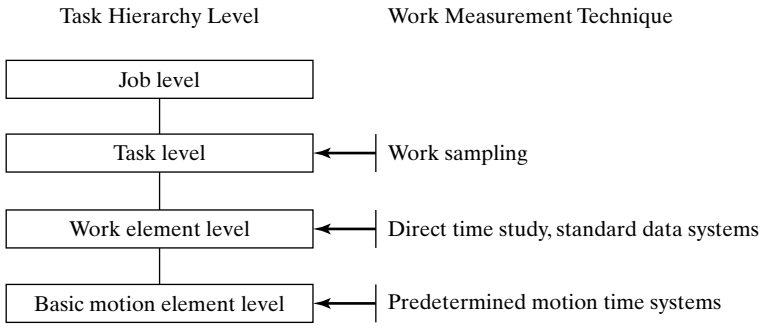


Figure 2 How the work measurement techniques correspond to different levels in the general task hierarchy.

[3]Some time study analysts attach an overall performance rating to the entire task while others prefer to rate the worker's performance during each work element.

$$T_n = T_{obs}(PR) \tag{1}$$

where T_n = normal time (the time required by a worker working at 100% pace for one cycle), min; T_{obs} = observed time, min; and PR = performance rating of the worker's pace observed by an analyst, usually recorded as a percentage during the observation but applied as a decimal fraction in the equation. If the task is repetitive, several work cycles are observed and the normalized time values are averaged to improve statistical accuracy.

To determine the standard time for the task, an allowance is added to the normal time to provide for personal time, fatigue, and delays. The calculation of the standard time is summarized in the following equation:

$$T_{std} = T_n(1 + A_{pfd}) \tag{2}$$

where T_{std} = standard time, min.; T_n = normal time, min.; and A_{pfd} = allowance factor for personal time, fatigue, and delays (PFD). The PFD allowance is explained in Section 3.1.

Predetermined Motion Time Systems. A predetermined motion time system (PMTS) relies on a database of basic motion elements (therbligs) such as reach, grasp, and move that are common to nearly all manual industrial tasks. Associated with each motion element is a set of normal times, whose values depend on the conditions under which the motion element was performed. For example, the normal time for a reach depends on the distance reached. Longer distances take more time. The normal time for a move depends on the distance moved and the weight of the object being moved. And so on.

To use a predetermined motion time system to set a standard time for a given task, the analyst lists all of the basic motion elements that comprise the task, noting their respective conditions, and retrieves the normal time for each element from the database. The normal times for the motion elements are then summed to obtain the normal time for the task. Finally, the standard time is calculated by adding a PFD allowance. This final step is the same as that expressed by equation (2) for direct time study. There are two advantages related to predetermined motion time systems: (1) performance rating is not required and (2) they can be applied to determine the time standard for a task before production.

Standard Data Systems. A standard data system (SDS) is a compilation of normal time values for work elements used in tasks that are performed in a given facility. These normal times are used to establish time standards for tasks that are composed of work elements similar to those in the database. An advantage of a standard data system is that a time standard can be set before the job is in production.

The normal time values in a standard data system are usually compiled from previous direct time studies, but they may be based on predetermined motion time data, work sampling data, or even historical time records. Large amounts of data are generally required to compile the database, and the work elements are performed under different work variables that may affect their normal time values. Accordingly, the normal time for a given element is a function of the work variable(s), and this functionality must be included in the database in the form of tables, charts, or mathematical equations.

To use a standard data system, the analyst first identifies the work elements that make up the task together with the values of the work variables respectively for each element. He or she then accesses the database to find the normal time for each element. The work element values are summed to determine the normal time for the task. As in the other work measurement techniques, an allowance is added to the normal time to compute the standard time.

Work Sampling. Work sampling uses random sampling techniques to study work situations so that the proportions of time spent in different activities can be estimated with a defined degree of statistical accuracy. Examples of the activities in a work sampling study might include setting up for production, producing parts, machine idle, and so on. The activities must be defined specifically for the work situation that the study is intended to address. The capability to include multiple subjects (e.g., workers, machines) in the study is one of the advantages of work sampling. Direct time study and predetermined motion time systems are generally limited to one worker for each study. A large number of observations over an extended period of time are usually made in a work sampling study in order to achieve the desired level of statistical accuracy. The period of the study must be representative of the activities normally performed by the subjects, and the observations must be made at random times in order to minimize bias; for example, if the workers knew when the observations would be made, it might influence their behavior.

The objectives in a work sampling study may be to measure machine utilization in a plant or to determine an appropriate allowance factor for use in setting standards in direct time study. The objective is not always to determine time standards. But when used to establish standards, the statistical errors inherent in the sampling procedure cause the "time standards" to be less accurate than those obtained by the other work measurement techniques.

Computerized Work Measurement. The purpose of work measurement and time standards is to improve the productivity of the workers who perform the value-adding tasks of the organization. But work measurement itself can be very time-consuming for the analysts who do it. A number of hardware and software products have been introduced commercially to improve the productivity of the analysts who perform work measurement. These products reduce the amount of time required by the analyst to set a time standard. Thus fewer analysts can set more standards.

Computerized products have been developed for all four of the work measurement techniques: direct time study, predetermined motion time systems, standard data systems, and work sampling. They make use of the ubiquitous personal computer (PC) as well as portable devices such as personal digital assistants (PDAs). In general, these products reduce the time and effort to perform work measurement by means of the following conveniences:

- Facilitating the collection of data at the work site in direct time study and work sampling
- Automatically performing the routine computations that previously had to be performed by the analyst

- Organizing the time standards files and databases
- Retrieving data from databases in predetermined motion time systems and standard data systems
- Assisting in the preparation of the documentation required in work measurement (e.g., methods descriptions, reports).

2 PREREQUISITES FOR VALID TIME STANDARDS

The time to perform one work cycle of a given manual task depends on the worker (his or her physical size and strength, as well as mental abilities), the worker's pace, the method used (hand and body motions, tooling, equipment, and work environment), and the work unit. As a prerequisite for establishing a standard time for a task, all of these factors must be standardized. The standardized factors are the following:

- The task is performed by an ***average worker***
- The worker's pace represents ***standard performance***
- The worker uses the ***standard method***
- The task is performed on a ***standard work unit*** that is defined before and after processing.

Establishing these definitions for the task is a requirement for setting a valid time standard, as depicted in Figure 3. And the definitions must be fully documented for future reference. If any of the factors change in future replications of the task, then it can be argued that the previously determined time standard is no longer valid. The only exception is the worker. Different workers are all expected to work by the same definition of standard time, whether they represent the "average worker" or not.

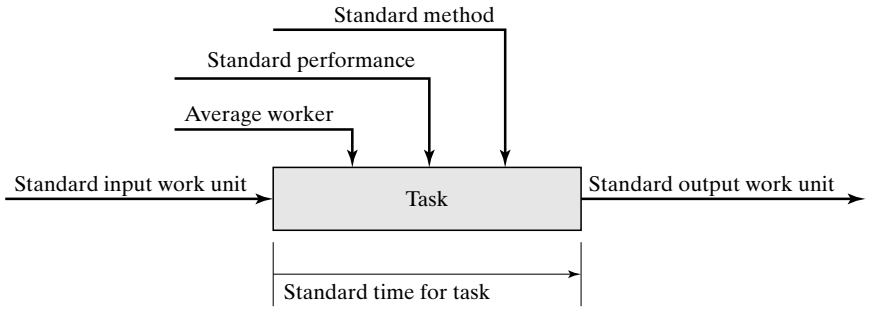


Figure 3 Model indicating the factors that must be standardized before a standard time can be set for a task.

2.1 Average Worker and Standard Performance

For the purpose of defining standard time, an ***average worker*** is a person who is representative of those who perform tasks similar to the task being measured. If similar tasks are performed mostly by men, and gender makes a difference in the ability to perform the task, then the average worker should be defined as a male. If performed mostly by women, then average worker means a female worker. Sometimes men are better suited to certain types of work, and sometimes women are better suited. In addition, the average worker is assumed to have learned the task and is practiced and proficient at it. He or she is well into the learning curve and is capable of performing the task consistently throughout the shift, which includes occasional rest breaks as befits the work situation.

To accomplish the task in the standard time, the average worker works at standard performance. Performance refers to the rate or speed with which a worker does the task. It is the pace of working. ***Standard performance*** is a pace of working that can be maintained by the average worker throughout an entire work shift without harmful effects on the worker's health or physical well-being. The work shift is usually considered to be an 8-hour workday during which the worker is allowed periodic rest breaks and may experience other interruptions. Standard performance is a pace the worker can achieve day after day. It is the normal pace of an average worker trained to perform the given task.

Several benchmarks of "standard performance" have been developed over the years. Two of the most popular are the following:

- Walking at 3 miles per hour (or 4.82 km per hour) on level flat ground
- Dealing four hands of cards from a 52 card deck in exactly 30 seconds.

The term ***normal performance*** is often used in place of standard performance. Basically, the two terms mean the same thing. Normal performance is 100% pace while the worker is working, while standard performance is 100% performance but with the proviso that periodic breaks are taken and other delays are likely to occur during the shift. A worker could not maintain 100% pace for 8 hours straight (imagine walking at 3 miles per hour for 8 straight hours, 24 miles, without an occasional rest stop). However, with periodic breaks a worker is able to work at 100% pace when he or she is working (with occasional rest breaks amounting to total of an hour or so, walking 24 miles in a day can be readily accomplished by a person who has trained for it[4]).

When an individual work cycle is performed at 100% performance, the time taken is the ***normal time*** for the cycle. The normal time is less than the standard time because the normal time does not include an allowance for time losses during the shift. When this allowance is added to the normal time, we have the ***standard time***. This is captured concisely in our equation for standard time, equation (2):

$$T_{std} = T_n (1 + A_{pfd})$$

[4]There are many accounts during the American Civil War (1861–1865) of entire armies marching distances more than 24 miles per day over terrain that was hardly flat and level.

The allowance factor A_{pfd} is intended to provide for the average time losses for personal needs, fatigue, and delays that occur during the shift. The allowance factor must be determined for the particular task or category of work to which it is applied. If the work is physically demanding, then a greater allowance factor must be provided for rest breaks. Allowances are discussed in Section 3.

Worker performance varies among individual workers. The variations occur due to age, sex, size and physical strength, physical conditioning, skill and aptitude for the task, experience and training, and motivation. For large numbers of workers, studies have shown that worker performance is normally distributed, and the ratio of the highest performing and lowest performing worker ranges between about 2.3 to 1.[5] The results (hypothetical) are presented in Figure 4. Worker performance in these studies is usually indicated by the hourly or daily production rate for a representative manual task.

It is common industrial practice to define standard performance as a pace that can be readily attained by the majority of workers. The companies (and unions representing the workers) want most workers to be able to achieve the standard performance fairly easily, so they define standard to be a performance that is less than the average in Figure 4. This is an additional condition on our preceding definition of standard performance. A typical multiplier is 1.30, although some companies prefer not to use a multiplier at all (in effect, the value of the multiplier for these companies is 1.0). A multiplier of 1.30 means that an average worker should be able to work at a pace that is 130% of the defined standard pace. This standard pace is sometimes referred to as ***day work pace***.

The production rate is equivalent to the reciprocal of the time to complete the task. Thus, for an output of 100 pieces per day (480 min) at average performance, the average task time would be 4.80 min (480 min/100 pc). Using a multiplier of 1.30 to define standard performance, this means that 100 pc/day is 130% of standard performance, and the corresponding standard time is therefore 4.80 (1.30) = 6.24 min, or a daily production rate of 77 pc. These values can be seen in Figure 5, a revision of our

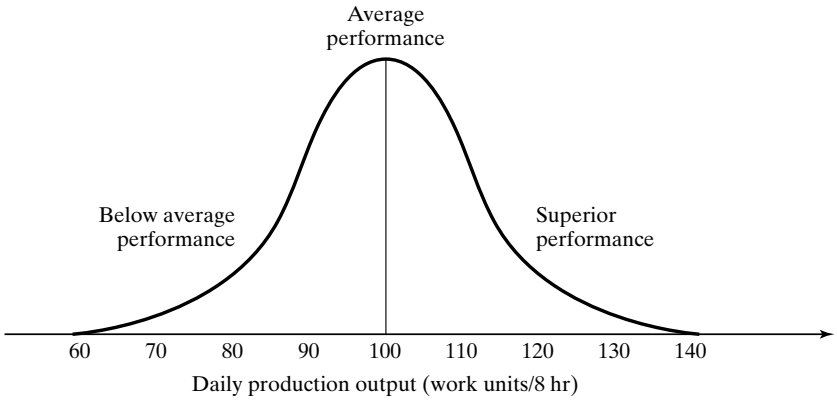


Figure 4 Distribution of performance (daily production output) for large numbers of workers.

[5]Several studies are cited in [8].

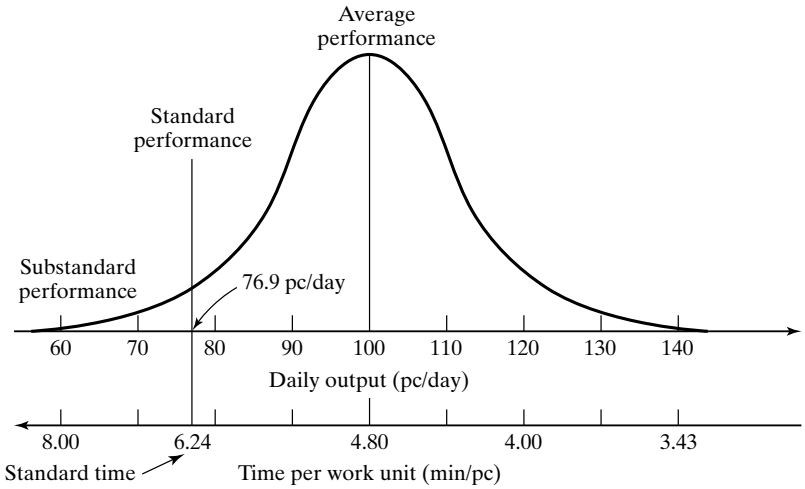


Figure 5 Distribution of performance and task time for large numbers of workers, indicating how standard time is typically defined as a time that can be readily achieved by most workers.

previous performance distribution. Accordingly, we can define ***standard performance*** in common industrial practice to be the pace of working at which the standard time for the task is 130% (or some other multiple, depending on company policy) of the mean time that can be maintained throughout an entire work shift by an average worker without harmful effects on the worker's health or physical well-being.

2.2 Standard Method

The ***standard method*** is the procedure that has been determined to be the optimum method for processing a work unit. The standard method satisfies the "one best method principle." It is the procedure that is the safest, quickest, most productive, and least stressful to the worker. It is constrained only by the practical conditions and economic limitations of the particular work environment. If the task is loading coal into a furnace in a steel mill, it is impractical and uneconomical to perform such work in a clean, air-conditioned environment that is void of safety hazards.

The standard method should include the following details about how a task is performed:

- *Procedure (actions and motions) used by the worker.* This is best described by listing the work elements that comprise the work cycle. The predetermined time systems provide a natural means for doing this because they list the basic motions that comprise the procedure. For direct time study and standard data systems, the procedure must be included in the statement of the standard method.
- *Tools.* This includes hand tools and portable power tools, fixtures, and gauges used in the procedure.

- *Equipment.* This includes machinery and safety equipment used to perform the task.
- *Workplace layout.* How is the workplace arranged? What are the locations of the parts, tools, etc., that will allow for the most efficient method? Where do the work units enter and exit the workplace? An overhead layout sketch of the workplace is a useful means of documentation.
- *Irregular work elements.* This includes tasks that are not performed every work cycle. Depending on the task, these elements may include changing tools as they wear out, changing parts containers as they become empty or full, and any periodic maintenance of the equipment that the worker is required to perform.
- *Working conditions.* Is the work performed outside or inside? What is the surrounding temperature, noise level, and other conditions that might affect the work environment?
- *Setup.* What setup of the physical tools and equipment is required to perform the task? How much time is allowed for the setup?

All of these factors should be thoroughly documented in a written statement of the standard method. At some future time, when the task is performed again after a long delay, the document will provide an invaluable and timesaving description of the task for which the time standard applies. If there is some dispute about the standard method or the standard time, the written statement will be a useful resource in resolving the dispute. Good documentation, a burden to prepare, is a blessing to practitioners.

2.3 Standard Input and Output Work Units

Most tasks involve a process or action performed on a work unit. The state or condition of the work unit is altered in some way by the task. Value is added to the work unit by performing the task on it (at least we hope so). The time required to accomplish the task is likely to depend on the condition of the input work unit. Therefore this condition must be specified as completely as possible in the standard method statement. Similarly, the condition of the output work unit must be specified. Exactly what changes have been made in the work unit as a result of the task performed on it? What is the final state of the completed work unit?

The starting and completed work units are usually easy to define in production because they deal with objects being processed. The status or condition of the starting work unit is usually defined explicitly by engineering documents that can be referenced in the standard method statement. The completed work unit is specified by similar documents. If an actual work part does not conform to these specifications, either before or after processing, then this nonconformance may necessitate a deviation from the standard method. For example, if a starting sand casting has too much metal to be machined due to variations or errors in the casting process, an additional machining pass across the surface may be required, which will take more time. This condition can be determined by comparing the actual part dimensions with those specified in the engineering drawing. If inspection reveals that the incoming part from the foundry is oversized beyond the tolerance, so that an extra machining pass is required, then additional time must be allowed for the extra pass (and the foundry should be notified to take corrective action for future deliveries).

In service work, the input and output work units are usually more difficult to define. However, most service tasks consist of a standard approach or method, which is the best procedure to accomplish the task. And the standard method may also have a "standard time" associated with it. An example is the routine automotive repair work performed in a car dealer's service department. Changing motor oil, replacing the oil filter, rotating four tires, and state inspection are all pretty standard jobs. For a given car model, each of these tasks has a standard method and a standard time. You can sometimes find the standard time on your bill from the service department.

3 ALLOWANCES IN TIME STANDARDS

In all of the work measurement techniques, the normal time is adjusted by an allowance factor to obtain the standard time. Allowances are used because there will be periods during the regular work shift when the worker is not working. The purpose of the allowance factor is to compensate for this lost time by providing a small increment of "allowance time" in each cycle. This way, even with the time losses, the operator will still be able to complete a fair day's work during the hours of the shift.

3.1 Accounting for Lost Time in the Workplace

There are various reasons why people do not work continuously during a regular shift. Some of the interruptions are work-related while others are not. Table 2 lists some typical causes of interruptions that occur in industrial work.

There is no denying that these interruptions occur during the work shift, and there is no denying that allowance time in some form should be provided to deal with them. The question is how much should be allowed, and how should it be provided to the worker. Companies address the allowance issue using two basic approaches, often in combination: (1) scheduled break periods during the shift and (2) a PFD allowance added to the normal time.

Scheduled Break Periods. Scheduled breaks are planned periods set aside during the shift as break time from work. Lunch breaks (or supper break for evening and night shifts) are almost always handled this way. Many companies treat rest breaks the same

TABLE 2 Typical Causes of Interruptions in the Workplace

Causes of Work-Related Interruptions	Reasons for Nonwork Related Interruptions
Machine breakdowns or malfunctions	Personal needs (e.g., restroom interruption)
Waiting for materials, parts, or other items necessary to proceed with work	Talking to coworkers about matters unrelated to work
Receiving instructions from foreman	Lunch break
Talking to coworkers about job-related matters	Smoke break
Waiting at a tool crib for the attendant to deliver a tool	Beverage break
Rest breaks to overcome fatigue for physiologically demanding work	Personal telephone call
Cleaning up at the end of the shift	

way. There is a specified rest break in the morning and one in the afternoon. The duration of these breaks is typically 5 to 15 minutes. All workers take their breaks during these specified times, and the workers are paid during these breaks.

In this approach, the shift is scheduled with breaks to divide the work day into periods of roughly equal length. For example, the day shift begins at 8:00 A.M. and ends at 4:30 P.M. There is a 15-minute rest break in midmorning, a lunch break from noon to 12:30 P.M., and another 15-minute rest break in midafternoon. Accordingly, the shift lasts 8.5 hours, which includes 8 hours of paid time (clock time).

The PFD Allowance. In this approach, an appropriate value of the ***personal time, fatigue, and delay*** (PFD) allowance factor (A_{pfd}) is determined for use in converting the normal time into the standard time. It results in extra time prorated in the time standard to compensate for the interruptions experienced by the worker during the shift, both work related and non-work related. Some companies include rest breaks in the allowance rather than using a scheduled break time for everyone as described above.

Personal time includes restroom breaks, phone calls, water fountain visits, and similar interruptions of a personal nature. A typical allowance for this category of interruption is 5%. A larger value would be appropriate if the work environment is hot and uncomfortable, and a lower value would be used for very favorable working conditions.

The ***fatigue*** or ***rest*** allowance is intended to compensate the worker for time that must be taken to overcome fatigue due to work-related stresses and conditions. For example, if the work is physiologically very demanding and the worker expends significant metabolic energy in performing it, then relaxation time should be allowed periodically for the body to recover. Some of the factors that cause fatigue in human workers are listed in Table 3. The allowance for fatigue for a given task is often determined by a combination of (1) the rest formulas and (2) negotiation between labor and management. The rest formulas provide a more or less scientific basis to make the determination. In practice, rest allowances are often the result of negotiated agreements between workers, perhaps represented by their union, and company management. An allowance of 5% is typical for fatigue in light and medium work. For heavy work it may be much higher, perhaps 20% or more.

Delays are unavoidable interruptions from work that occur at random times during the day. They usually refer to work-related events, such as, some of the interruptions listed in Table 2. It should be mentioned that work delays associated with a worker's personal needs (e.g., a sudden urge for a cigarette break) should not be counted in this

TABLE 3 Factors Causing Fatigue in Workers

Physical Factors	Mental and Cognitive Factors	Environmental and Work Factors
Standing	Concentration and attention	Poor lighting
Abnormal body position	Mental strain	Noise
Use of force	Monotony and tediousness	Fumes
Expenditure of muscular energy	Eyestrain	Heat
		Atmospheric conditions

Source: Adapted from a publication by the International Labour Office, *Introduction to Work Study* (1964) and other sources, as cited in [1].

category of interruption, since they are already included in the personal needs category. The delay category is generally reserved for delays that are ultimately the responsibility of management. For example, a machine breakdown may seem like a random occurrence, but management is responsible for the maintenance of the machine so that breakdowns are avoided. Other examples include waiting for parts or tools, waiting for an elevator, and waiting for the foreman's instructions. In general, these kinds of work-related delays must be controlled by company management. Certainly the worker is not responsible.

Determining the delay allowance is more challenging than determining allowances for personal needs or fatigue. There are traditions and/or formulas, often adjusted by negotiations, for determining these other allowances. But delays are usually random events that vary from day to day, and determining an allowance factor to account for them boils down to a problem of collecting data in the workplace so that an average lost-time value can be found. Two techniques are used to collect the data: (1) intensive observation over several days and (2) work sampling. Of the two methods, work sampling is usually preferred for several reasons: (1) it can be conducted over a longer period of time (e.g., weeks instead of days), so the plant activity observed is more likely to represent normal operations; (2) it can include a much larger scope of plant operations, because the observer is not limited to one location; (3) it does not require continuous observation, so it is more comfortable for workers and observer; and (4) statistical measures can be calculated to assess the accuracy of the results derived from the data.

The three allowances for personal needs, fatigue, and delays are then combined into a single allowance factor, which we symbolize A_{pfd}.

3.2 Other Types of Allowances

In addition to the PFD allowance factor, there are other reasons for adding allowances to the standard time of a task. They are not as common as the PFD allowance, but in some work situations they are quite appropriate. These other allowances are applied in addition to A_{pfd}, not as a substitute for it.

Contingency Allowances. The contingencies for which these allowances are provided are usually because of some kind of problem with the task or the production equipment used to perform it. Some examples of these problems are listed in Table 4. The

TABLE 4 Examples of Problems for Which Contingency Allowances Might Be Provided

Problem Area	Problems and Examples
Materials or parts	Starting materials or parts are out of specification, and extra time is needed to correct the nonconformance (e.g., oversized casting that requires an extra machining pass or slower feed rate).
Process	The manufacturing process is not in statistical control (too much variability), and additional time is required to inspect every piece rather than inspect on a sampling basis.
Equipment	Equipment is malfunctioning or breaking down more frequently than what is provided by the unavoidable delay factor, and additional time is needed to compensate the worker to make adjustments, lubricate the machine more frequently, or other extra task(s) not included in the standard time.

problems are beyond what is covered by the regular PFD allowance, and so an additional allowance is provided. Contingency allowances should not be greater than 5% [1]. The problems that contingency allowances are intended to address should not occur in the first place, but if they occur, the problem itself should be solved. Contingency allowances may be needed on a temporary basis to keep production going, but the long-term solution is to fix the underlying problem.

Policy Allowances. These allowances are intended to cover special work situations that are usually associated with a wage incentive system. An example is the ***machine allowance*** that is added to the machine-paced portion of a work cycle in the operation of a worker-machine system. The machine allowance provides an opportunity for the worker to maintain a high rate of earnings even though he or she has control over only a portion of the cycle. It is company policy that the worker be given this opportunity. The application of the machine allowance to compute the standard time is as follows:

$$T_{std} = T_{nw}(1 + A_{pfd}) + T_m(1 + A_m) \tag{3}$$

where T_{nw} = normal time of the worker during the worker-controlled portion of the cycle, min; A_{pfd} = PFD allowance; T_m = machine cycle time, min; and A_m = machine allowance. If company policy does not recognize the need for a separate machine allowance when computing the time standard, then either $A_m = 0$ or it is set equal to the value of the PFD allowance. The following example illustrates the way the worker benefits from a machine allowance.

Example 1 Use of Machine Allowance in a Wage Incentive Plan

A wage incentive plan pays workers a daily wage at a rate of \$15.00/hr multiplied by the number of standard hours accomplished during the shift. One worker-machine task in the plant includes worker-paced elements totaling a normal time of 1.00 min and machine-paced elements with a time of 3.00 min. The PFD allowance is 15%. Determine the standard time for the task given that (a) $A_m = 0$ and (b) $A_m = 30\%$. (c) What does a worker earn for the day under each policy if he or she produces 115 parts that day?

Solution: **(a)** $T_{std} = 1.00\,(1 + 0.15) + 3.00\,(1 + 0) = 1.15 + 3.00 = 4.15$ min.

(b) $T_{std} = 1.00\,(1 + 0.15) + 3.00\,(1 + 0.30) = 1.15 + 3.90 = 5.05$ min.

(c) For the standard of 4.15 min, the number of standard hours is $H_{std} = 115(4.15)/60 = 7.95$ hr and the worker's earnings would be \$15.00 (7.95) = \$119.25.[6]

For $T_{std} = 5.05$ min, $H_{std} = 115(5.05)/60 = 9.68$ hr and the worker's earnings would be \$15.00(9.68) = \$145.19.

Comment: Which is fair? A PFD allowance of 15% equates to a lost time per day of 62.6 min in an 8-hour shift, which leaves an actual working time of 417.4 min. Of this time, machine time took 115 cycles x 3.00 min/cycle = 345 min, which leaves 417.4 – 345 = 72.4

[6]Most incentive plans pay a guaranteed base rate, so the worker's wage for the day would be 8(\$15.00) = \$120.00.

min for worker time. Given that 115 cycles were completed by the worker, his performance during the worker-paced portion of the cycle must have been 115(1.00)/72.4 = 1.59 = 159%.

Certainly this level of performance would seem to merit some bonus above the base rate of \$15/hr. The standard time in (a), T_{std} = 4.15 min, does not even pay the worker \$15/hr (unless there is a guaranteed base rate, see footnote). The standard time in (b), T_{std} = 5.05 min, does earn a respectable bonus of \$30.19. The worker's efficiency is 9.68/8.0 = 1.21 or 121%.

The other side of the argument is that the worker has free time for 345 min during the shift (because this is machine cycle time), so why should he or she be paid a premium bonus for the entire shift for only working 72.4 min of total shift time? This is a difficult issue, one on which common interest can often be found by both labor and management in keeping the machinery operating as high a proportion of time as possible (a win-win situation for both labor and management). Labor wins by earning higher bonuses. Management wins by using the equipment more efficiently. ■

Other types of policy allowances include ***training allowances*** for workers whose responsibilities include teaching other workers their jobs, and ***learning allowances*** for workers who are learning a new job or new employees who are just beginning work. In nearly all of the situations covered by policy allowances, a worker's earnings would be adversely affected if the allowances were not applied. Workers would be reluctant to train others or to learn new jobs unless some form of compensation were provided to cover the losses in earnings they would otherwise suffer for doing so. If a wage incentive plan is not used, there is less reason to have policy allowances since the worker's wage is not affected. The reason they are called policy allowances is because they are based on company policy, rather than any time interruptions actually experienced during the work shift.

4 ACCURACY, PRECISION, AND APPLICATION SPEED RATIO IN WORK MEASUREMENT

Work measurement is a measurement process. There are similarities and differences between the measurement of work and the measurement of physical variables in science and technology. Let us begin by defining measurement as it is applied in the physical sciences. ***Measurement*** is a procedure in which an unknown quantity is compared to a known standard, using an accepted and consistent system of units. The measurement provides a numerical value of the quantity of interest, within certain limits of accuracy and precision. All measurement systems are based on seven basic physical quantities: (1) length, (2) mass, (3) time, (4) electrical current, (5) temperature, (6) luminous intensity, and (7) matter. All other physical quantities (e.g., area, volume, velocity, force, electrical voltage) are derived from these seven basic quantities.

Time is one of the seven basic physical quantities, and time is the common quantity of interest in the measurement of work. The standard unit for time is the second.[7] Although time standards in work measurement often use alternative units (e.g., minutes, hours), they all can be related back to seconds.

[7]The ***second*** is defined in the System International as the "duration of 9,192,631,770 cycles of the radiation associated with a change in energy level of the cesium atom."

Three important attributes of a measurement system are accuracy, precision, and speed of response. These attributes are discussed in the following sections, and comparisons are made between the measurement of quantities in science and the measurement of work.

4.1 Accuracy and Precision

Measurement ***accuracy*** is the degree to which the measured value agrees with the true value of the quantity of interest. The measurement system is accurate if it is free of ***systematic errors***, which are positive or negative deviations from the true value that are consistent from measurement to measurement.

Measurement ***precision*** refers to the repeatability of the measurement system. High precision means that the random errors in the measurement procedure are small. It is generally assumed that random errors follow a normal statistical distribution with mean (μ) equal to zero and standard deviation (σ) indicating the amount of dispersion in the measured data. The normal distribution has well-defined properties, one of which is that 99.73% of the population is included within $\pm 3\sigma$, and this is often used as a quantitative indicator of the measurement system's precision.

The distinction between accuracy and precision is depicted in the three normal distributions of Figure 6. In distribution (a), the random errors in the measurement are large (large variance), which indicates that the precision is low, but the mean value agrees with the true value of the variable of interest, which indicates perfect accuracy. In (b), the random errors are low (high precision), but the measured value differs from the true value (low accuracy). In (c), we have high accuracy and high precision.

Accuracy and precision are important in work measurement, but these terms that are so readily applied to physical measurements are fraught with difficulty when applied to the measurement of work, especially direct time study. Accuracy is concerned with closeness to the true value, but how can that definition be applied to the time that should be allowed a worker to perform a given task? What is the true value of a task time? Measurement is a procedure in which an unknown quantity is compared to a known standard. In work measurement, there is no way to make that comparison because the closest that we can come to a known standard is in the minds of the time study analysts.

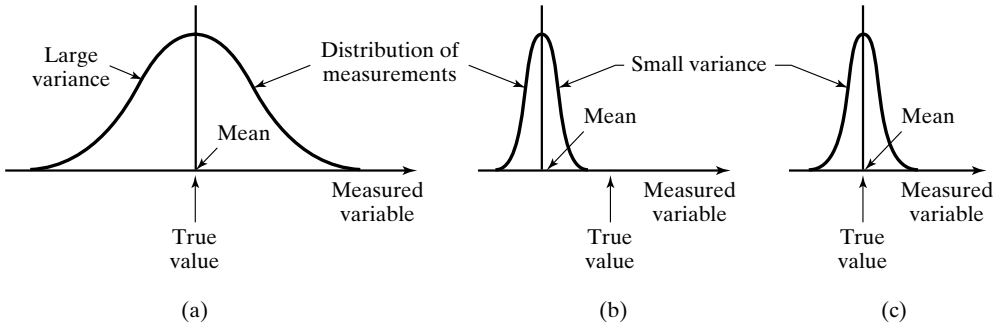


Figure 6 Accuracy versus precision in measurement: (a) high accuracy but low precision, (b) low accuracy but high precision, and (c) high accuracy and high precision.

And that standard is not based on time; it is based on a definition of performance (the organization's definition of standard performance). The issue becomes more confounded in direct time study because of performance rating. The time study analyst records the observed time of a given work element and then applies a performance rating to that time in order to align the observed time with the analyst's concept of standard performance.[8] We must conclude that accuracy in work measurement is elusive because it is impossible to scientifically determine the true value of the quantity of interest.

Although there are difficulties in determining accuracy in work measurement, the precision of the time standard resulting from a work measurement study can be determined, and this is sometimes used as a pseudonym for accuracy. The precision of the time standard is determined at a certain reliability level. For example, we state that the standard time for the task is 4.00 min, and that we are 95% certain that the actual time is within ± 5% of that time. In other words, in 95 out of 100 time studies performed on the task, the resulting standard time values will lie between 3.80 min and 4.20 min. This is analogous to confidence intervals in statistics, and we make use of them in Chapter 13 to determine how many cycles must be observed in direct time study in order to achieve a known level of reliability and precision in the time standard. The term ***consistency*** also applies here, and it refers to the variations in standard time values among different time study analysts who study the same task. Good consistency means small variations in the resulting time standards by different analysts. The term ***precision*** refers to the expected variability within a single time study or the time standard resulting from that study.

The relative accuracies of the various methods of determining time standards are of interest. These are summarized in Figure 7. Estimation is likely to be the least accurate because it is not based on any actual data or measurement. It relies solely on the estimator's judgment, which may not be very good. By comparison, historical records provide hard data on the average time required for the previous production order, and that is generally an improvement over judgment. Nevertheless, the accuracy of the record is sometimes questionable. Were the parts counted correctly, or did workers inflate

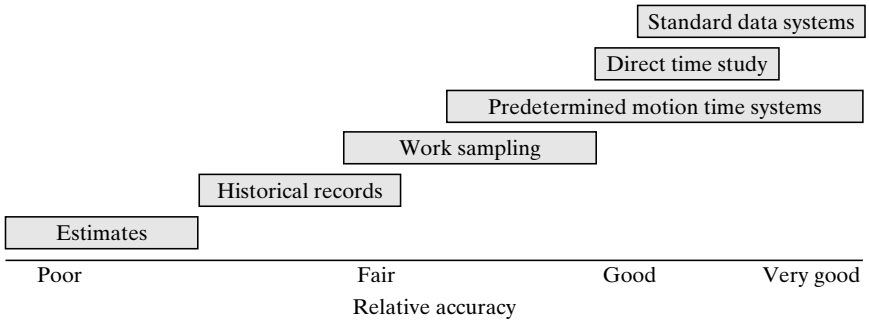


Figure 7 Methods of determining time standards ranked by relative accuracy.

[8]Mitchell Fein, a leading industrial engineering consultant for many years, made the following observation about direct time study: "No other measurement system which tests data for statistical validity alters the observed data" [4].

numbers to make themselves look more productive? Did the time sheet provide an accurate accounting of time spent on the particular order, or was other work done and charged to this order? How long did the setup take for this order? Although historical records are likely to be more accurate than pure estimation, they suffer from the fact that they are not based on a measurement of the actual work; instead, they are based on measures of the work output and the time that was recorded for that output.

In terms of relative accuracy, the work measurement techniques are more accurate than estimation or historical records. Of the four work measurement techniques, work sampling is not as accurate for determining time standards as the other three techniques. This due to (1) the statistical errors that result from the sampling process and (2) the absence of any attempt to analyze and improve the method used in the task.

4.2 Application Speed Ratio

Accuracy and precision are not the only criteria in selecting a work measurement technique. The amount of time required to perform the work measurement procedure is also important. In other measurements, the ***speed of response*** refers to the lag between the time when the measuring device is applied and when the measured value is available to the user. We would like to obtain our measurement of the variable of interest as quickly as possible, whether we are measuring a physical variable in science or a task time in a work situation.

In work measurement, the related term is ***application speed ratio***, which is the ratio of the time required to determine a time standard relative to the standard time itself. For example, an application speed ratio of 100 means that it takes 100 min of analyst time to determine a 1.0-min time standard. In general there is an inverse relationship between application speed ratio and "accuracy." Methods for setting time standards that can be applied quickly tend to be less accurate than those that take more time to apply.

The time required to perform any of the work measurement techniques is significantly greater than the time required to estimate a task time or to compute the task time from historical records. If accuracy of the task time is not important, or if the work consists of a variety of "odd jobs," where it would be impractical to set a time standard for each job, then estimation or historical records are preferable as a way of developing the time values. Figure 8 indicates the relative application speeds of the various methods for setting time standards. The PMT systems have a large range of application speeds

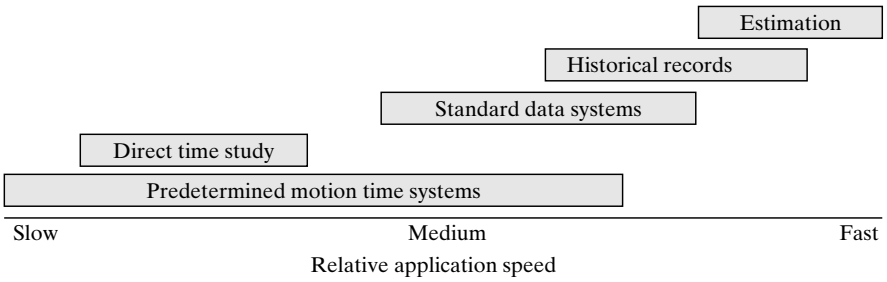


Figure 8 Methods of determining time standards ranked by relative application speed.

because of the variety of systems available, some of which are very accurate but very time consuming, while others are less accurate and take less time. The application speeds for standard data systems and historical records are relatively fast, but these systems require a substantial investment of time to develop before they are operational. Work sampling is omitted from the chart because it requires an extended period of data collection before standards can be determined.

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REVIEW QUESTIONS

1 What is the standard time for a task? Provide a definition.

2 What are some of the characteristics typical of industrial situations in which time standards would be beneficial?

3 What are the functions and applications of accurately established time standards in an organization?

4 Identify the three basic methods to determine time standards.
5 What are the four basic work measurement techniques?
6 What is the difference between the normal time and the standard time for a task?
7 How does computerized work measurement reduce the time and effort of the time study analyst?
8 What are the prerequisites for valid time standards?
9 Identify the two common benchmarks of standard performance that are often used.
10 What are the details that should be included when defining the standard method?
11 What is a PFD allowance in time standards?
12 What are some of the reasons why workers experience lost time during a work shift?
13 What is a contingency allowance?
14 What is the difference between accuracy and precision in a measurement system?
15 Why is accuracy an elusive quality in work measurement?
16 What is precision in work measurement?
17 What is consistency in work measurement?
18 What is the application speed ratio in work measurement?

PROBLEMS

Determining Time Standards

1 The average observed time for a repetitive work cycle in a direct time study was 3.27 min. The worker was performance rated by an analyst at 90%. The company uses a PFD allowance factor of 13%. What is the standard time for this task?
2 A time study analyst observed a repetitive work cycle performed by a worker and then studied the task using a predetermined motion time system. During the observation, the analyst made a record of the work elements. He also made a note of the worker's performance and rated it as 85%. Back in the office, the analyst listed the basic motion elements for the task and retrieved from his computer the corresponding normal time values. The sum of the normal times for the basic motion elements was 1.38 min. The company uses a PFD allowance factor of 15%. Determine the standard time for the task.
3 The ABC Company uses a standard data system to set time standards. One of the time study analysts listed the three work elements for a new task to be performed in the shop and then determined the normal time values to be 0.73 min, 2.56 min, and 1.01 min. The company uses a PFD allowance factor of 16%. Determine the standard time for the task.

Allowances

4 In the ABC machine shop, workers punch in at 8:00 A.M. and punch out at 4:30 P.M. The labor-management agreement allows 30 min for lunch, which is not counted as part of the 8-hour shift. In determining the allowance for computing time standards, two 15 min breaks are provided (personal time and fatigue), one in the morning and one in the afternoon; and 20 min have been negotiated as lost time due to supervisor interruptions and equipment malfunctions. What PFD allowance factor should be added to the normalized time to account for these losses in the computation of a standard time, so that if workers work at standard performance, they will produce exactly 8 standard hours?

5 In the WS&FP plant, workers punch in at 8:00 A.M. and punch out at 5:00 P.M. The labor-management agreement allows 1 hour for lunch, which is not counted as part of the 8-hour shift. In determining the allowance for computing time standards, two 12 min breaks are included (personal time and fatigue), one in the morning and one in the afternoon; and 35 min are included as lost time due to interruptions and delays. What PFD allowance factor should be added to the normalized time to account for these losses in the computation of a standard time, so that if workers work at standard performance, they will earn exactly 8 standard hours?

6 Determine the PFD allowance to be used for computing time standards in the following situation. Second shift workers punch in at 3:30 P.M. and punch out at 12:00 midnight. They are provided 30 min for supper at 6:00 P.M., which is not counted as part of the 8-hour shift. For purposes of determining the allowance, 30 min of break time (personal time and fatigue) are allowed each worker. In addition, the plant allows 35 min for lost time due to unavoidable delays. What should the PFD allowance factor be?

7 The work shift at the ABC Company runs from 7:30 A.M. to 4:15 P.M. with a 45 min break for lunch from 11:30 to 12:15 P.M. that does not count as part of the work shift (workers are not paid for this time). The company provides two 12-min rest breaks during working hours (paid time), one in the morning and one in the afternoon. The company also allows 25 min per day for personal needs (paid time). In addition, a work sampling study has shown that on average, unavoidable delays in the plant result in 20 min lost time per worker per day (paid time). Determine the PFD allowance factor for the following two management policies on allowances: (a) the two 12-min breaks are both scheduled breaks that all workers take at the same time and (b) the two 12-min breaks are included in the allowance factor so that workers can take their breaks whenever they please.

Direct Time Study

1 Direct Time Study Procedure
  1.1 Define and Document the Standard Method
  1.2 Divide the Task into Work Elements
  1.3 Time the Work Elements
  1.4 Rate the Worker's Performance
  1.5 Apply Allowance to Compute Standard Time
2 Number of Work Cycles to Be Timed
3 Performance Rating
4 Time Study Equipment
  4.1 Stopwatch Time Study
  4.2 Video Cameras

Direct time study, also known as ***stopwatch time study,*** involves the direct and continuous observation of a task using a stopwatch or other timekeeping device to record the time taken to accomplish a task.[1] While observing and recording the time, an appraisal of the worker's performance level is made. These data are then used to compute a standard time for the task, adding an allowance for personal time, fatigue, and delays.

Direct time study was the first work measurement technique to be used, dating back to around 1883. It is inextricably connected with the origins of industrial engineering. Computerized techniques have improved the accuracy and application speed of direct time study as well as the database management functions that support it.

The use of direct time study is most appropriate for tasks that involve a repetitive work cycle, at least a portion of which is manual. This kind of work is common in

[1]The Work Measurement and Methods Standards Committee (ANSI Standard Z94.11-1989) defines time study as follows: "A work measurement technique consisting of careful time measurement of the task with a time measuring instrument, adjusted for any observed variation from normal effort or pace and to allow adequate time for such items as foreign elements, unavoidable or machine delays, rest to overcome fatigue, and personal needs. Learning or progress effects may also be considered. If the task is of sufficient length, it is normally broken down into short, relatively homogeneous work elements, each of which is treated separately as well as in combination with the rest."

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batch and mass manufacturing. Also, standards for routine office work can often be established using direct time study. Performing a direct time study is time-consuming and is therefore best justified when the job will have a relatively long production run, and/or there will be repeated orders in the future. A limitation of this work measurement technique is that it cannot be used to set a time standard prior to the start of production.

1 DIRECT TIME STUDY PROCEDURE

The procedure for determining the standard time for a task using direct time study can be summarized in the following five steps:

1. Define and document the standard method.
2. Divide the task into work elements.
3. Time the work elements to obtain the observed time for the task.
4. Evaluate the worker's pace relative to standard performance, a procedure called ***performance rating***. This is used to determine the normal time. Steps 3 and 4 are accomplished simultaneously.
5. Apply an allowance to the normal time to compute the standard time.

Steps 1 and 2 are preliminary steps before actual timing begins, during which the analyst becomes familiar with the task and attempts to improve the work procedure before defining the standard method. In steps 3 and 4, several work cycles are timed, each one performance rated independently. Finally, the values collected in steps 3 and 4 are averaged to determine the normalized time. An appropriate allowance factor for the kind of work involved is then added to compute the standard time for the task. Let us examine each of the steps in more detail.

1.1 Define and Document the Standard Method

Before defining and documenting the standard method, a methods engineering study should be undertaken to ensure that the standard method obeys the "one best method" principle—the best method that can be devised under the present economic and technological circumstances. All of the steps in the method should be defined. Any special tools, gauges, or equipment that can improve the task should be designed and included in the method. If there are irregular elements in the work cycle, the frequency with which these elements are to be performed should be stated explicitly. If the labor-management climate in the facility allows, the worker's advice and opinion should be sought in developing the standard method. Once the standard method has been defined, it should be difficult or impossible for the operator to make further improvements.

The standard method should be thoroughly documented. The company should have forms and/or checklists to make certain that all information about the method is included. An example of a methods description form is shown in Figure 1. The documentation should enumerate details about the procedure (hand and body motions), tools, equipment and the machine settings used for the equipment (e.g., feeds and speeds

Direct Time Study

Date	**Standard Method Description for Direct Time Study**	Page of
Operation	Dept.	Part No.
Machine	Analyst	
Methods Improvements (check if implemented)	Sketch of Workplace:	

Work Element No. and Description with Machine Parameters for Machine Cycles	Freq.	Tools and Gauges

Additional Notes

Figure 1 Form for documenting the standard method.

on machine tools), workplace layout, irregular work elements, working conditions, and setup. If there is a work unit associated with the task, then it should also be specified, both its starting and its completed conditions. A videotape (or other video medium) of the standard method can also be made as part of the documentation. This is especially helpful for complex tasks in which details of the method are difficult to explain in writing.

There are several reasons why thorough documentation of the standard method is important:

- *Batch production*. If the task is associated with batch production, then it is likely to be repeated at some time in the future. The time lapse may be significant between the previous batch and the next batch. A different worker may be assigned to perform the task on the next batch. The statement of the standard method provides the worker and the foreman with a complete description of the task and the procedure for doing it, as well as any tooling or equipment needed. This avoids "reinventing the wheel."
- *Methods improvement by the operator*. At some future time, the operator may discover a way to improve the method.[2] A question then arises: Should the methods improvement be incorporated into a new standard method (which would require a change in the standard time), or should the operator be allowed to benefit from the improvement without formally changing the standard method? The typical labor-management contract only allows for retiming the task if it can be demonstrated that a real change in the method has been made. If the operator has indeed made a methods improvement, this can be identified by comparing the method in use with the standard method description.
- *Disputes about the method*. If the operator complains that the standard for the task is too tight, or there is some other reason for a dispute about the standard method, documentation of the standard method can be used to settle the dispute.[3] Perhaps the operator is using a less efficient method than the standard method, or a certain hand tool that should be used in the operation has been neglected. These problems can be addressed by reference to the standard method documentation.
- *Data for standard data system*. Time standards developed by direct time study are sometimes used in standard data systems. Good documentation of the standard method, especially regarding the work elements and associated normal times, is essential in developing the database for a standard data system.

[2]This possibility seems at odds with the fact that the one best method has already been devised and has been defined as the standard method. However, the intelligence and ingenuity of the worker should not be underestimated. It must be acknowledged that the worker may be able to figure out some motion shortcut or hand tool or other gimmick that reduces the time for the work cycle, even though the work cycle has been subjected to a thorough methods study.

[3]A "tight" standard means that the standard time is too short, making it difficult or impossible to accomplish the task within the time standard. A "loose" standard is one in which the time is relatively long, making it easy to achieve a high worker efficiency.

1.2 Divide the Task into Work Elements

Any task can be divided into work elements. A ***work element*** is a series of motion activities that are logically grouped together because they have a unified purpose in the task. The description of the standard method can be organized into work elements suitable for use in direct time study. Indeed, the most natural way to describe the standard method is often as a list of work elements. Some practical guidelines for defining work elements in direct time study are presented in Table 1. An important reason for defining the

TABLE 1 Guidelines for Defining the Work Elements in Direct Time Study

Guideline	Explanation and Examples
Each work element should consist of a logical group of motion elements.	The work element should have a unified purpose, such as reaching for an object and moving it to a new location (e.g., reach, grasp, move, and place). There would be no purpose in separating the reach from the move motions since they both involve the same object.
Beginning point of one element should be end point of preceding element.	There should be no gap between one element and the next in the task sequence. Otherwise, the time of the gap is omitted from the recorded total time.
Each element should have a readily identifiable end point.	A readily identifiable end point can be easily detected during the study. It can often be anticipated to allow reading of the watch more conveniently. An audible sound, such as the actuation of a pneumatic device, provides a readily identifiable end point.
Work elements should not be too long.	If a work element is very long (i.e., several minutes), it should probably be divided into multiple elements that are timed separately. Machine semiautomatic cycle time is an exception. Some machine cycles can take several minutes and should be identified as one element.
Work elements should not be too short.	A practical lower limit in direct time study is around 3 sec. Below this, reading accuracy may suffer. If a video camera is used for timing purposes, shorter elements may be possible.
Irregular work elements should be identified and distinguished from regular elements.	Irregular elements are work elements that do not occur every cycle. The frequency with which they should be performed must be noted. The time(s) for the irregular element(s) are prorated across the regular work cycle when the standard time is computed.
Manual elements should be separated from machine elements.	Manual elements depend on the operator's performance (pace) and therefore vary over time. Machine elements are generally constant values that depend on machine settings. Once the settings are established, the actuation time shows no perceptible variation.
Internal elements should be separated from external elements.	Internal elements are performed by the operator during the machine cycle. In most cases, they do not affect the overall work cycle time. External elements are performed outside of the machine cycle. They contribute to the overall work cycle time.

Source: Adapted from guidelines in [13] and other sources.

work elements is that the worker may exhibit different performance levels on different elements. Accordingly, these performance levels are rated and recorded separately by the time study analyst.

1.3 Time the Work Elements

Once the work elements have been defined, the analyst is ready to collect data. The time data are usually recorded on a time study form, similar to the one shown in Figure 2. Space is provided for a listing of the work elements, which can be referenced to the more complete standard method documentation. Each element should be timed over several cycles to obtain a reliable average, and the form is designed for recording multiple cycles of the task. The appropriate number of cycles can be determined using the statistical techniques described in Section 2. For the convenience of the analyst, the time study form is usually held in a special clipboard that also holds the stopwatch used in the study.

There are several pieces of equipment that can be used to record the times for the work cycles. We describe the products in Section 4. The traditional instrument in direct time study is the stopwatch, which is usually calibrated in decimal minutes. There are two principal methods for using a stopwatch in direct time study: (1) snapback timing method and (2) continuous timing method. In the ***snapback timing method***, the watch is started at the beginning of every work element by snapping it back to zero at the end of the previous element. The reader must therefore note and record the final time for that element just as the watch is being zeroed. (Most stopwatches, especially electronic watches, have features that facilitate this reading.) In the ***continuous timing method***, the watch is zeroed at the beginning of the first cycle and allowed to run continuously throughout the duration of the study. The analyst records the running time on the stopwatch at the end of each respective element. Some analysts prefer to adapt the continuous method by zeroing at the beginning of each work cycle, so that the starting time of any given work cycle is always zero. This facilitates cycle-to-cycle comparisons during the study.

There are two advantages to the snapback method: (1) the analyst can readily see how the element times vary from one cycle to the next, and (2) no subtraction is necessary, as in the continuous timing method, to obtain individual element times. The advantages of the continuous method include the following: (1) when the clock is continuously running, elements are not as easily omitted by mistake, (2) regular and irregular elements can be more readily distinguished, and (3) not as much manipulation of the stopwatch is required as in the snapback method.

1.4 Rate the Worker's Performance

While observing and recording the time data, the analyst must simultaneously observe the performance of the worker and rate this performance relative to the definition of standard performance used by the organization. Other terms for performance include pace, speed, effort, and tempo. Standard performance is given a rating of 100%. A performance rating greater than 100% means that the worker's performance is better than standard (which results in a shorter observed work cycle time), and less than 100% means poorer performance than standard (and a longer observed time).

Direct Time Study

Date	**Direct Time Study Observation Form**			Page of
Operation		Dept.	Part No.	
Machine			Tooling	
Worker			Worker No.	
Analyst	Start Time	Finish Time	Elapsed Time	

Work Elements, Machine Settings, and Observations				Cycle No. (regular elements)										
Element Number and Description	Feed	Speed		1	2	3	4	5	6	7	8	9	10	Avg T_n
1			T_{obs}											
			PR											
			T_n											
2			T_{obs}											
			PR											
			T_n											
3			T_{obs}											
			PR											
			T_n											
4			T_{obs}											
			PR											
			T_n											
5			T_{obs}											
			PR											
			T_n											
6			T_{obs}											
			PR											
			T_n											
7			T_{obs}											
			PR											
			T_n											
8			T_{obs}											
			PR											
			T_n											
Normal time = Sum of T_n (regular work elements)														

Irregular Element and Description	Freq	T_0	T_f	PR	T_n		**Calculation of Standard Time T_{std}**	
A							Sum of T_n (regular work elements)	
B							Sum of freq x T_n (irregular elements)	
C							Total T_n per cycle	
D							PFD allowance A_{pfd}	
E							Standard time $T_{std} = T_n(1 + A_{pfd})$	
Additional Notes								

Figure 2 Direct time study form.

The observed time is subsequently multiplied by the performance rating to obtain the ***normal time*** (other names include ***normalized time*** and ***base time***) for the element or cycle. The calculation is summarized in the following equation:

$$T_n = T_{obs}(PR) \tag{1}$$

where T_n = normal time, min; T_{obs} = observed time, min; and PR = performance rating, usually expressed as a percentage but used in the equation as a decimal fraction. The symbols T_n and T_{obs} can be used to represent individual work elements or the entire work cycle, depending on how the data are taken and recorded.

Performance rating is the most difficult and controversial step in direct time study. The reason is that it requires the judgment of the analyst to assess the value of PR. The analyst's judgment of standard performance may differ from that of the worker who is being observed. It is in the worker's interest and advantage to be rated at a high performance level during the study, because that will mean that the normal time and ultimately the standard time for the task will be longer (resulting in a looser standard). Thus, it will be easier for the worker to achieve a higher efficiency level as the job continues. This is especially important to the worker if he or she is paid on a wage incentive plan. We consider performance rating and the issues surrounding it in Section 3.

1.5 Apply Allowance to Compute Standard Time

To obtain the standard time for the task, a PFD allowance is added to the normal time, as calculated in the following equation:

$$T_{std} = T_n(1 + A_{pfd}) \tag{2}$$

where T_{std} = standard time, min; T_n = normal time, min; and A_{pfd} = allowance factor for personal time, fatigue, and delays. This is often expressed as a percentage but used as a decimal fraction in our equation. The function of the allowance factor is to inflate the value of the standard time relative to the normal time in order to account for the various reasons why the operator loses time during the work shift. The allowance factor represents an average for the type of work, equipment, and conditions under which the operator works. Some days the worker may lose more time, other days less time, than what is provided by the allowance factor. In the long run it is intended to average out to a realistic value.

Example 1 Determining a Standard Time for Pure Manual Work

A direct time study was taken on a manual work cycle using the snapback timing method. The regular work cycle consisted of three elements, identified as *a*, *b*, and *c* in the following table. Element *d* is an irregular element performed every 5 cycles. Observed times and performance ratings of the elements are also given. Determine (a) the normal time and (b) the standard time for the work cycle, using an allowance factor of 15%.

Work Element	*a*	*b*	*c*	*d*
Observed time	0.56 min	0.25 min	0.50 min	1.10 min
Performance rating	100%	80%	110%	100%

Solution **(a)** The normal time for the cycle is obtained by multiplying the observed element times by their respective performance ratings and summing. In the case of element *d*, this normal time is prorated over 5 cycles.

$$T_n = 0.56(1.00) + 0.25(0.80) + 0.50(1.10) + 1.10(1.00)/5$$
$$= 0.56 + 0.20 + 0.55 + 0.22 = 1.53 \text{ min}$$

(b) The standard time is computed by adding the allowance.

$$T_{std} = 1.53(1 + 0.15) = 1.76 \text{ min.}$$ ■

If the task includes a machine cycle, then company policy may provide for a machine allowance factor to be included in the standard time computation. The standard time equation becomes

$$T_{std} = T_{nw}(1+A_{pfd})+T_m(1+A_m) \tag{3}$$

where T_{nw} = normal time of the worker external elements, min; T_m = machine cycle time, min; A_m = machine allowance factor; and the other terms have the same meanings as before. If the company does not include a separate machine allowance, then $A_m = 0$ in the equation, or it uses the regular allowance A_{pfd}. Equation (3) assumes that there are no internal elements in the cycle. If the work cycle includes internal elements, then it must be determined whether the sum of the worker internal elements or the machine cycle time is larger in order to determine the normal time and the standard time.

Example 2 Determining a Standard Time for a Task That Includes a Machine Cycle

The snapback timing method was used in a direct time study of a task that includes a machine cycle. Elements *a*, *b*, *c*, and *d* are performed by the operator, and element *m* is a machine semiautomatic cycle. Element *b* is an internal element performed simultaneously with element *m*, and element *d* is an irregular element performed once every 15 cycles. Observed times and performance ratings are given in the table below. The PFD allowance factor is 15%, and the machine allowance is 20%. Determine (a) the normal time and (b) the standard time for the work cycle.

Worker element	*a*	*b*	*c*	*d*
Observed time, manual	0.22 min	0.65 min	0.47 min	0.75 min
Performance rating	100%	80%	100%	100%
Machine element		*m*		
Observed time, machine	(idle)	1.56 min	(idle)	(idle)

Solution **(a)** The normal time must take account of which element, *b* or *m*, has the larger value. Also, element *d* must be prorated across 15 cycles.

$$T_n = 0.22(1.00) + \text{Max}\{0.65(0.80), 1.56\} + 0.47(1.00) + 0.75(1.00)/15$$
$$= 0.22 + 1.56 + 0.47 + 0.05 = 2.30 \text{ min}$$

(b) The same comparison between elements *b* and *m* must be made in computing the standard time.

$$T_{std} = (0.22 + 0.47 + 0.05)(1 + 0.15) + \text{Max}\{0.52(1 + 0.15), 1.56(1 + 0.20)\}$$
$$= 0.85 + 1.87 = 2.72 \text{ min}$$ ■

2 NUMBER OF WORK CYCLES TO BE TIMED

One of the practical issues in taking a time study is determining how many work cycles should be timed. The reason this issue arises is that there is statistical variation in the times of respective elements from one work cycle to the next. Direct time study involves a sampling procedure, and the objective is to determine a value for the population work element time as accurately as is possible and practical. As we increase the sample size, we expect the accuracy of the estimate to improve. On the other hand, increasing the sample size also increases the cost of taking the time study. It seems reasonable to try to find a balance between these competing factors.

There is inherent variability in any human activity, and performing manual work is a human activity. Work element times vary from cycle to cycle because of the following reasons:

- Variations in hand and body motions
- Variations in the placement and location of parts and tools used in the cycle
- Variations in the quality of the starting work units (e.g., a plastic molded part with flash that must be trimmed)
- Mistakes by operator (e.g., operator accidentally drops the work part)
- Errors in timing the work elements by the analyst
- Variations in worker pace

All of these variations are manifested in the work element times recorded for the cycle. Performance rating is supposed to compensate for the last item, variations in worker pace. However, because performance rating requires judgment by the analyst, that also introduces error and variation.

For analysis purposes, we assume that the observed work element times are normally distributed about the true value of the work element time. For practical purposes, we identify the longest work element in the cycle, or the most critical element (the one in whose accuracy we are most interested), as the element to focus on. Let us call this element time T_e. Our objective is to be able to identify the true value of T_e within a certain confidence interval. For example, we might state that we want to be 95% confident that the true value of T_e lies within ±10% of the observed average value of the element time. Let us identify the average value simply as $\bar{x}$. This is illustrated in Figure 3, which shows the distribution of observed time values taken during the time study.

The area under the normal curve between ±10% of $\bar{x}$ represents the probability that the true value of element time T_e lies within ±10% of $\bar{x}$. The general statement of the confidence interval can be expressed as follows:

$$\Pr\left(T_e \text{ lies within } \bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = (1 - \alpha) \tag{4}$$

where $z_{\alpha/2}$ = standard normal variate, $\mathtt{s}$ = standard deviation of the population element time, n = sample size, and $(1 - \mathtt{a})$ = confidence level. The term $z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ is made equal to our desired interval size (e.g., ±10% of $\bar{x}$), and we solve for the value of n, the number of work cycles to time.

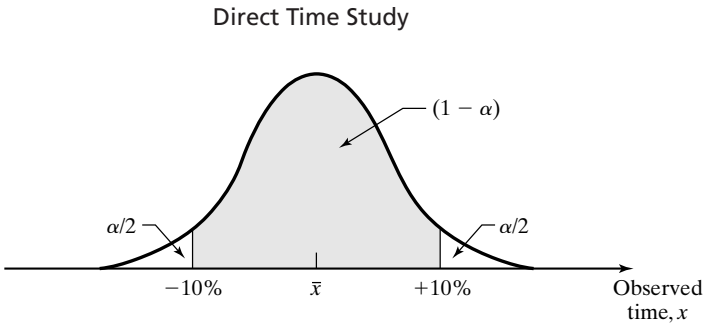


Figure 3 Distribution of observed element times in a direct time study.

There are two difficulties with the preceding analysis: (1) we do not know the population standard deviation s and (2) the sample size n that we are usually dealing with is relatively small. The standard deviation must be estimated from the sample itself, just as the mean $\bar{x}$ must be determined from the sample. The value of the sample standard deviation is given by

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} \tag{5}$$

where s = sample standard deviation, x = individual values of the observed times collected during the study, and the other terms have previously been defined. Because the sample size n is usually small (e.g., less than 30), the sample values are distributed according to the student t rather than the normal distribution.

Taking these adjustments into account, we can make the following confidence interval statement to replace equation (4):

$$\Pr\left(T_e \text{ lies within } \bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}\right) = (1-\alpha) \tag{6}$$

Our objective is to find the value of sample size n that will satisfy our specification of α and interval size for the values of $\bar{x}$ and s that have been determined from the data collected. Let us express the interval size in more general terms:

$$\text{Interval size} = \bar{x} \pm k\bar{x}$$

where k = a proportion that specifies the interval size (e.g., $k = 10\%$ or 0.10). We will set that equal to $\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$, from which the following can be established:

$$k\bar{x} = t_{\alpha/2}\frac{s}{\sqrt{n}} \tag{7}$$

Rearranging and solving for n, we have

$$n = \left(\frac{t_{\alpha/2}s}{k\bar{x}}\right)^2 \tag{8}$$

where values of $t_{\alpha/2}$ can be found in the Appendix: Statistical Tables.

Example 3 Determining the Number of Work Cycles to Be Timed

A time study analyst has collected 10 readings on a particular work element of interest and would like to consider how many more cycles to time. Based on the sample, the mean time for the element is $\bar{x} = 0.40$ min and the sample standard deviation $s = 0.07$ min. At a 95% confidence level, how many cycles should be timed to ensure the actual element time is within ±10% of the mean?

Solution: We have 10 – 1 = 9 degrees of freedom in the t distribution, so the value of $t_{\alpha/2}$ at the 95% confidence level ($\alpha/2 = 0.025$) is 2.262. Using equation (8), we obtain

$$n = \left(\frac{2.262(0.07)}{0.10(0.40)} \right)^2 = 15.7$$

This should be rounded up to 16 cycles total. Since 10 cycles have already been timed, the analyst needs data from 6 more work cycles. ■

The time study analyst may wish to turn the problem around; that is, to determine the size of the confidence interval at a certain desired confidence level, given that the observed times have already been recorded for a completed time study. This can be done using equation (8) but rearranging to solve for k.

Notwithstanding the preceding statistical approach, the problem of determining how many work cycles should be timed often boils down to the practical issue of how much time can we afford to spend doing the time study. Certain observations are germane here:

- The statistical accuracy of the time data increases when more observations are taken. The equations tell us that. If the task being studied is an important one in the factory operations, then more time should be spent on the time study to ensure the highest possible accuracy of the standard. More time should be spent studying high production operations than low production operations.
- Shorter work cycles allow more cycles to be timed during the study. For work cycles under 1 minute, 30 or more cycles are practical. For longer work cycles, fewer cycles can be included. For work cycles of 30 minutes, it may only be practical to observe six work cycles (which is 3 hours or more of time study observation).
- It is a judgment call by the time study department as to the appropriate number of work cycles to be timed. The judgment attempts to balance the desire for statistical accuracy against the cost of the time study.

3 PERFORMANCE RATING

Performance rating, also called ***performance leveling***, is the step in direct time study in which the analyst observes the worker's performance and records a value representing that performance relative to the analyst's concept of standard performance. A number of different performance rating methods have been proposed over the years, but the

simplest and most common method is based on speed or pace.[4] It is sometimes referred to as speed rating. The other methods are rarely used today.

As its name indicates, ***speed rating*** involves rating the operator's pace of working relative to the concept of standard performance. A rating greater than 100% means that the worker is working faster than standard pace, and less than 100% means the worker is slower than standard. Pace is the only factor that is rated. However, pace depends on the type of work involved. To rate the operator's pace, the analyst must use judgment that is based on previous training and experience in rating similar work. An analyst who is experienced in machine shop time studies would find this experience of limited value if suddenly faced with the job of conducting time studies in a garment factory. The concept of standard performance is different for different kinds of work. Standard performance for a highly repetitive manual assembly task in which the cycle time is 1 minute should be defined differently than standard performance in a design prototype modeling shop in which the worker must apply technical skill and artistic talent to the job. And within a given category of work, some tasks are more difficult than others. Accordingly, the analyst must observe two aspects of the work scene during performance rating: (1) the degree of difficulty of the task and what 100% pace would be for that task, and (2) the subject worker's pace relative to the concept of 100% pace.

Let us examine these two aspects of performance using one of our accepted definitions of standard performance, walking at 3 miles/hr (4.83 km/hr). This is a 20-minute mile. The more complete definition is walking at 3 miles/hr on level flat ground, using 27-inch steps.[5] Also, it is assumed that the walker carries no load. If any of these conditions are not satisfied (i.e., level ground, 27-inch steps, no load), then it represents a change in the degree of difficulty, and so the definition of 100% performance might need adjusting. For example, walking uphill on a 3% slope might have a standard pace of only 2.5 miles/hr. But if the conditions of the definition are satisfied, so that standard performance is 3 miles/hr, then an individual's walking speed can be measured to assess his or her performance rating relative to this standard. In fact, walking is a situation where we can very precisely determine performance level if we assume that performance level is directly proportional to the speed of the walker. For example, suppose the walker is able to complete 1 mile in 18 min (0.30 hr) instead of 20. This is a speed of 3.33 miles/hr (1 mile ÷ 0.3 hr). Compared to 3.0 miles/hr, the performance level is 111%. We do not necessarily know that the walker's performance was a constant 111% (3.33 miles/hr) throughout the entire mile, but we know that on average it was 111%.

Unfortunately, very few work situations lend themselves to such a precise measurement of performance. Thus, performance rating comes down to a matter of the analyst judging the worker's pace relative to his or her concept of standard performance for the type of work involved. Direct time study is not the only instance where judgment is used to rate people. There are many areas of human activity in which judgments are used by experts to decide the relative performance of subjects, and the expert's judgments are readily accepted. Table 2 presents a list of activities in which judges rate the

[4]For the interested reader, the other performance rating methods are described in [14].
[5]This definition was apparently proposed by Ralph Presgrave, one of the early workers in time and motion study.

TABLE 2 Activities in Which Trained and Experienced Judges Rate the Contestants

Human Activities That Are Judged	Other Activities That Are Judged
Figure skating	Dog shows
Diving events	Farm shows
Gymnastics	Gourmet food contests
Bodybuilding contests	Art contests
Beauty contests	High school science fair contests
Umpire calling balls and strikes behind home plate in a baseball game	Grading of (college) freshman English essays

contestants, often reducing the judgment to a numerical score. Just as in these other areas, time study analysts must be trained and become practiced in performance rating. Training is accomplished through the use of training films (videotapes) that show scenes of workers performing various kinds of industrial tasks at a variety of performance levels. The workers' pace in these scenes have been performance rated by experts, so that a trainee can judge each scene and compare his or her ratings with those of the experts. On-the-job practice is obtained by accompanying experienced analysts during actual time studies.

Depending on the type of work and length of the work cycle, the analyst must decide whether to rate the performance of individual elements (called ***elemental rating***) or to rate the entire cycle (***overall rating***). If the work cycle is relatively short (e.g., less than a minute) and the work content is similar throughout the cycle, then convenience may favor rating the entire cycle. If the cycle is longer and involves different kinds of work elements, on which the operator exhibits different pace levels, then elemental rating is more appropriate.

Operator selection can influence the performance rating procedure. The pace observed by the analyst depends to some extent on the worker's skill, experience, exertion level, and attitude toward time study. If there is more than one worker performing the same task that is to be time studied, then it is in the analyst's interest to select a skilled worker who is familiar with the job (has had time to learn the task) and who accepts time study as a necessary management tool. Such workers are usually willing to cooperate with the time study analyst, even to the extent that they will attempt to work at a pace consistent with the company's definition of standard performance, thus relieving the analyst of being forced to attach a rating that is significantly different from 100%. It has been found that when the subject performs at a pace that is within about 15% of standard, the resulting standard time value is much more likely to be fair and accurate and accepted by the worker. If the subject's performance during observation deviates significantly from standard, not only does this make life more difficult for the analyst, it increases the likelihood that the standard will not be an accurate measure of the task.

Potential difficulties are encountered in performance rating because a potential conflict of interest exists between the subject worker and the time study analyst. It is always in the worker's interest for the performance rating to be high due to its effect on

the computation of standard time. Notwithstanding the difficulties, we can identify the following characteristics of a well-implemented performance rating system:

- *Consistency among tasks.* The performance rating system should provide consistent ratings from one task to another. A worker who is able to perform at 125% efficiency on one task should be able to achieve the same efficiency level on any other task. The performance rating used to establish the normal time is a key factor in obtaining this consistency among tasks.
- *Consistency among analysts.* The performance rating for a task should not depend on which time study analyst does the rating. Consistency within a group of time study analysts can be measured using the deviations of individual *PR* values about the group average. Niebel and Freivalds indicate that a deviation of $\pm 5\%$ is considered adequate [14]. Proper initial training and periodic refresher training for the analysts in a company should provide this level of consistency.
- *Easily understood.* The rating system should be easy to explain by the analyst and simple to understand by the worker.
- *Related to standard performance.* The company should have a well-defined concept of standard performance. To the extent possible, this concept should be documented for reference by workers. Time study analysts should apply this definition when performance rating a task.
- *Machine-paced elements rated at 100%.* The operator has no control over machine-paced elements and therefore deserves no performance rating for these elements. Any adjustment for machine-cycle elements should be done using a machine allowance when computing the standard time.
- *Performance rating recorded during observation of task.* The performance should not be rated and recorded afterward.
- *Worker notification.* At the conclusion of the time study observation session, the analyst should inform the worker of the performance rating that was observed. This sets a proper tone for the relationship between the two people. It puts pressure on the analyst to be as fair as possible in the rating process, and it allows the worker to express agreement or disagreement with the rating.

4 TIME STUDY EQUIPMENT

In direct time study, the analyst directly observes the task as a worker performs it. The equipment used by the analyst ranges from simple to sophisticated. In our coverage we divide the range into three categories: (1) traditional time study using a stopwatch, (2) video camera to record the observation on tape, and (3) computerized time study techniques.

4.1 Stopwatch Time Study

The traditional equipment used in direct time study consists of a stopwatch and a time study form on which to record the times during observation. There are alternatives to the stopwatch, such as a wristwatch or wall clock, but the proper professional instrument

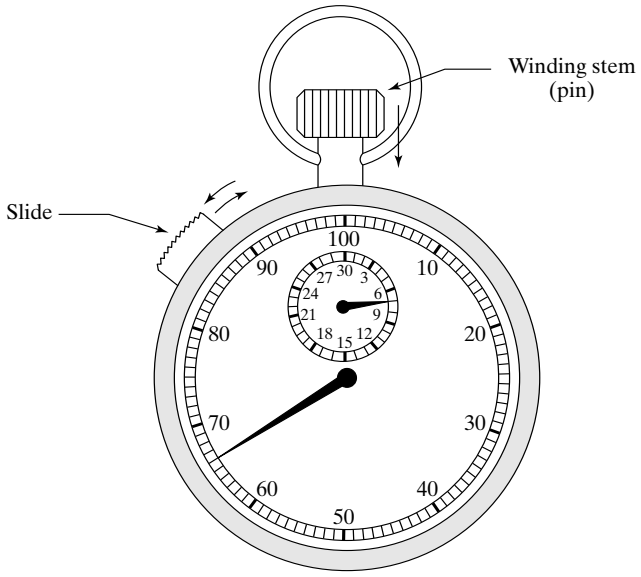


Figure 4 Mechanical stopwatch calibrated in decimal minutes (0.01 min) on the large dial. The small dial reads up to 30 minutes. Depressing the pin (also used as the winding stem) zeros the watch. The slide is used to start and stop the timing of individual work elements.

is a stopwatch. The time study form is usually held on a clipboard, which is designed to hold the stopwatch as well. Following the observation, the collected time data are analyzed and a time standard is calculated for the task.

Stopwatches can be classified as mechanical or electronic. ***Mechanical stopwatches*** have the familiar round face with one or more rotating hands, as shown in Figure 4. They are designed to be held in the time study clipboard or the palm of one's hand so the fingers can actuate the pin and slide at the top. Mechanical stopwatches are not a recent development. They have been used for time study since the late 1800s. Modern stopwatches for time study are typically graduated in one of three time measurement scales: (1) decimal minutes, shown in Figure 4, (2) decimal hours, and (3) TMUs (Time Measurement Units); 1 TMU = 0.00001 hr or 0.036 sec). The reason for having different scales is that different organizations have adopted different units to express their time standards. Decimal minute watches seem to be the most common, decimal hours less common, and TMUs are sometimes used because they are usually the time units in certain predetermined motion time systems.

Electronic stopwatches (Figure 5) provide a digital display of the time; otherwise their operating features are similar to those of mechanical watches. They have largely replaced mechanical stopwatches in time study for several compelling reasons. Because they display a digital readout, they are easier to read than the graduated mechanical dial, especially if the mechanical hand is moving. Because they are easier to read, reading errors occur less frequently. They are lighter in weight and generally less

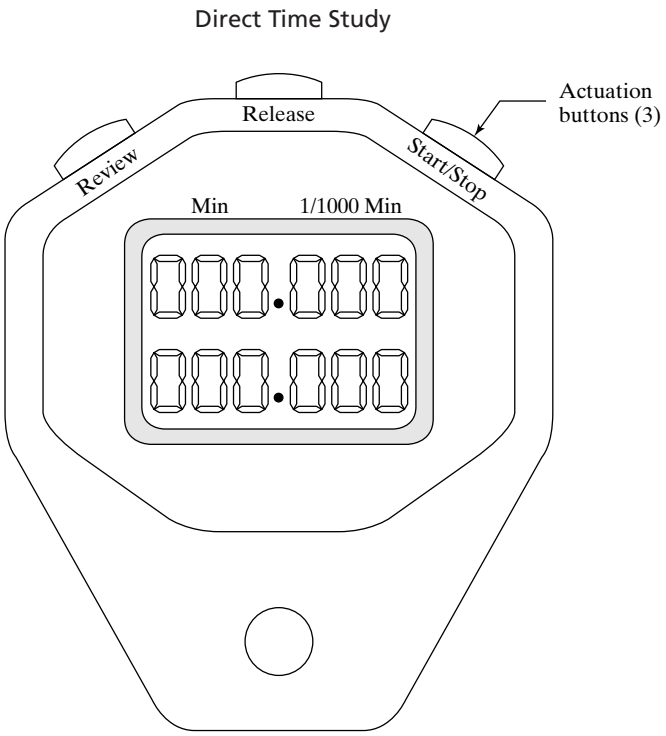


Figure 5 Electronic stopwatch with LED read-out, that can be used for continuous or snapback timing methods.

susceptible to damage when dropped or otherwise abused. Electronic watches are also more accurate and precise, which allows them to be used to read shorter work elements more conveniently. Some electronic watches can be switched back and forth between different time scales, unlike their mechanical counterparts. They can be used for either the continuous timing mode or the snapback mode. Finally, electronic stopwatches are less expensive than mechanical stopwatches.

The direct time study observation form (Figure 2) provides space for the analyst to record the observed times and the performance rating of the operator. Either the continuous timing method or the snapback method can be used. Observed times for multiple work cycles can be entered on the form.

4.2 Video Cameras

Video cameras have become a familiar piece of electronic equipment not only to consumers but to time study analysts as well. They are useful in direct time study because they provide a complete visual and audio record of the method used by the worker. The method is captured in much greater detail than the analyst can observe with the naked eye or is willing to include in any written documentation of the method. The tape also provides an accurate record of the times taken by each work element (accurate to 1/30 sec based on a frame-to-frame frequency of 30 Hz in the United States, or 1/25 sec based on 25 Hz in Europe). The tape can be subsequently played and replayed (in slow motion if desired) to

analyze the method, perhaps uncovering possible improvements in the task. Work element times can be analyzed for cycle-to-cycle variations. Operator performance can be rated in a much more relaxed and objective setting than what is possible in the shop floor environment. And finally, a standard time can be determined for the task from the tape.

Later, if a dispute arises about the standard time, the method, or the performance rating, the tape can be viewed to help settle the dispute, perhaps avoiding a formal grievance. If the operator wants to watch his own work performance, perhaps to decide whether he agrees with the rating, this can be accommodated. The tape can be used for training purposes for new time study analysts. The videotape provides an objective and accurate record of the worker performing the task using the method for which the time standard was determined.

Video cameras can be used for a variety of work situations in motion and time study: short repetitive work cycles, long cycles with variable elements, worker-machine systems, crews of workers, and several workers working independently but captured in the field of vision of the camera.[6]

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REVIEW QUESTIONS

1 Define direct time study.
2 Identify the five steps in the direct time study procedure.
3 Why is it so important to define and document the standard method as precisely and thoroughly as possible?
4 What is the snapback timing method when using a stopwatch during direct time study?
5 What is the continuous timing method when using a stopwatch during direct time study?
6 Why is performance rating a necessary step in direct time study?
7 Why is an allowance added to the normal time to compute the standard time?
8 What are some of the causes of variability in the observed work element times that occur from cycle to cycle?
9 Why is the student *t* distribution rather than the normal distribution used in the calculation of the number of work cycles to be timed?
10 What is the difference between elemental performance rating and overall performance rating?
11 What are the characteristics of a well-implemented performance rating system?
12 What are the advantages of electronic stopwatches compared to mechanical stopwatches?

PROBLEMS

Note: Some of the problems in this set require the use of parameters and equations.

Determining Standard Times for Pure Manual Tasks

1 The observed average time in a direct time study was 2.40 min for a repetitive work cycle. The worker's performance was rated at 110% on all cycles. The personal time, fatigue, and delay (PFD) allowance for this work is 12%. Determine (a) the normal time and (b) the standard time for the cycle.

2 The observed element times and performance ratings collected in a direct time study are indicated in the table below. The snapback timing method was used. The PFD allowance in the plant is 14%. All elements are regular elements in the work cycle. Determine (a) the normal time and (b) the standard time for the cycle.

Work element	*a*	*b*	*c*	*d*
Observed time (min)	0.22	0.41	0.30	0.37
Performance rating(%)	90	120	100	90

3 The standard time is to be established for a manual work cycle by direct time study. The observed time for the cycle averaged 4.80 min. The worker's performance was rated at 90% on all cycles observed. After eight cycles, the worker must exchange parts containers, which

took 1.60 min, rated at 120%. The PFD allowance for this class of work is 15%. Determine (a) the normal time, (b) the standard time for the cycle, and (c) the worker's efficiency if the worker produces 123 work units during an 8-hour shift.

4 The snapback timing method was used to obtain the average times and performance ratings for work elements in a manual repetitive task. (See table below). All elements are worker-controlled and were performance rated at 80%. Element *e* is an irregular element performed every five cycles. A 15% allowance for personal time, fatigue, and delays is applied to the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 120% on all manual elements for 7 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

Work element	*a*	*b*	*c*	*d*	*e*
Observed time (min)	0.32	0.85	0.48	0.55	1.50

5 The continuous timing method is used to direct time study a manual task cycle consisting of four elements: *a*, *b*, *c*, and *d*. Two parts are produced each cycle. Element *d* is an irregular element performed once every six cycles. All elements were performance rated at 90%. The PFD allowance is 11%. Determine (a) the normalized time for the cycle, (b) the standard time per part, (c) the worker's efficiency if the worker completes 844 parts in an 8-hour shift during which she works 7 hours and 10 min.

Element	*a*	*b*	*c*	*d*
Observed time (min)	0.35	0.60	0.86	1.46

6 The readings in the table below were taken by the snapback timing method of direct time study to produce a certain subassembly. The task was performance rated at 85%. In addition to the above regular elements, an irregular element must be included in the standard: each rack holds 20 mechanism plates and has universal wheels for easy movement. After completing 20 subassemblies, the operator must move the rack (which now holds the subassemblies) to the aisle and then move a new empty rack into position at the workstation. This irregular element was timed at 2.90 min and the operator was performance rated at 80%. The PFD allowance is 15%. Determine (a) the normalized time for the cycle, (b) the standard time, and (c) the number of parts produced by the operator, if he or she works at standard performance for a total of 6 hr and 57 min during the shift.

Element and Description	Observed Time (min)
1. Pick up mechanism plate from rack and place in fixture.	0.42
2. Assemble motor and fasteners to front side of plate.	0.28
3. Move to other side of plate.	0.11
4. Assemble two brackets to plate.	0.56
5. Assemble hub mechanism to brackets.	0.33
6. Remove plate from fixture and place in rack.	0.40

7 The time and performance rating values in the table below were obtained using the snapback timing method on the work elements in a certain manual repetitive task. All elements are worker-controlled and were performance rated at 85%. Element *e* is an irregular element performed every five cycles. A 15% PFD allowance is applied to the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance

during actual production is 125% on all manual elements for 7 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

Work element	*a*	*b*	*c*	*d*	*e*
Observed time (min)	0.61	0.42	0.76	0.55	1.10

Determining Standard Times for Worker-Machine Tasks

8 The snapback timing method was used to obtain average times for work elements in one work cycle. The times are given in the table below. Element *d* is a machine-controlled element and the time is constant. Elements *a*, *b*, *c*, *e*, and *f* are operator-controlled and were performance rated at 80%; however, elements *e* and *f* are performed during the machine-controlled element *d*. The machine allowance is zero (no extra time is added to the machine cycle), and the PFD allowance is 14%. Determine (a) the normal time for the cycle and (b) the standard time for the cycle.

Element	*a*	*b*	*c*	*d*	*e*	*f*
Observed time (min)	0.24	0.30	0.17	0.76	0.26	0.14

9 The continuous timing method in direct time study was used to obtain the element times for a worker-machine task as indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, *e*, and *f* are operator-controlled and external to the machine cycle, and they were performance rated at 80%. If the machine allowance is 25%, and the worker PFD allowance is 15%, determine (a) the normal time, (b) the standard time for the cycle, and (c) the worker's efficiency if the worker completes 360 work units working 7.2 hr on an 8-hour shift.

Element	*a*	*b*	*c*	*d*	*e*	*f*
Observed time (min)	0.18	0.30	0.88	1.12	1.55	1.80

10 A worker-machine cycle is direct time studied using the continuous timing method. One part is produced each cycle. The cycle consists of five elements: *a*, *b*, *c*, *d*, and *e*. Elements *a*, *c*, *d*, and *e* are manual elements, external to machine element *b*. Every 16 cycles the worker must replace the parts container, which was observed to take 2.0 min during the time study. All worker elements were performance rated at 80%. The PFD allowance is 16%, and the machine allowance is 20%. Determine (a) the normalized time for the cycle, (b) the standard time per part, and (c) the worker's efficiency if the worker completes 220 parts in an 8-hour shift during which he or she works 7 hr and 12 min.

Element	Description	Cumulative Observed Time (min)
a	Worker loads machine and starts automatic cycle.	0.25
b	Machine is engaged in automatic cycle.	1.50
c	Worker unloads machine.	1.75
d	Worker files part to size.	2.30
e	Worker deposits part in container.	2.40

11 In the preceding problem, a recommendation has been submitted for elements *d* and *e* to be performed as internal elements (accomplished simultaneously) with machine element *b*. The worker would file the part from the previous cycle and deposit it in the container while the current part is being processed in the machine automatic cycle. Performance rating and allowances are the same as in the previous problem. Determine (a) the normal time for the cycle, (b) the standard time per part, and (c) the number of parts that will be produced if the worker's efficiency is 115% and he works a total of 7 hr and 12 min during an 8-hour shift.

12 The snapback method was used to time study a worker-machine cycle consisting of elements *a*, *b*, and *c*. Elements *a* and *b* are worker-controlled and were performance rated at 100% during the time study. Element *c* is machine-controlled. Elements *b* and *c* are performed simultaneously. The PFD allowance is 12% and the machine allowance is 10%. One work piece is produced each cycle. Determine (a) the normal time, (b) the standard time for the cycle. (c) the number of pieces completed if the worker works 7 hr and 10 min during an 8-hour shift, and his performance level is 135%.

Element	*a*	*b*	*c*
Observed time (min)	1.25	0.90	0.80

13 The snapback timing method in direct time study was used to obtain the times for a worker-machine task. The recorded times are listed in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, and *d* are operator-controlled and were performance rated at 90%. Elements *a* and *b* are external to machine-controlled element *c*. Element *d* is internal to the machine element. The machine allowance is zero, and the PFD allowance is 13%. Determine (a) the normal time and (b) the standard time for the cycle. The worker's actual time spent working during an 8-hour shift was 7.08 hours, and he produced 420 units of output during this time. Determine (c) the worker's performance during the operator-controlled portions of the cycle and (d) the worker's efficiency during this shift.

Element	*a*	*b*	*c*	*d*
Observed time (min)	0.34	0.25	0.68	0.45

14 The table below lists the average work element times obtained in a direct time study using the snapback timing method. Elements *a* and *b* are operator-controlled. Element *c* is a machine-controlled element and its time is constant. Element *d* is a worker-controlled irregular element performed every five cycles. Elements *a*, *b* and *d* were performance rated at 80%. The worker is idle during element *c*, and the machine is idle during elements *a*, *b*, and *d*. One product unit is produced each cycle. To compute the standard, no machine allowance is applied to element *c*, and a 15% PFD allowance is applied to elements *a*, *b*, and *d*. (a) Determine the standard time for this cycle. (b) If the worker produces 220 units on an 8-hour shift during which 7.5 hours were actually worked, what was the worker's efficiency? (c) For the 220 units in (b), what was the worker's performance during the operator-paced portion of the cycle?

Element	*a*	*b*	*c*	*d*
Observed time (min)	0.60	0.45	1.50	0.75

15 The continuous timing method in direct time study was used to obtain the times for a worker-machine task as indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, *e*, and *f* are operator-controlled and were performance rated at 95%; they are all external elements performed in sequence with machine element *c*. The machine allowance is 30%, and the PFD allowance is 15%. Determine (a) the normal time and (b) standard time for the cycle. (c) If the operator works at 100% of standard performance in production and one part is produced each cycle, how many parts are produced if the total time worked during an 8-hour day is 7.25 hours? (d) For the number of parts computed in (c), what is the worker's efficiency for this shift?

Element	*a*	*b*	*c*	*d*	*e*	*f*
Observed time (min)	0.22	0.40	1.08	1.29	1.75	2.10

16 The snapback timing method was used to obtain average time and performance rating values for the work elements in a certain repetitive task. The values are given in the table below. Elements *a*, *b*, and *c* are worker-controlled. Element *d* is a machine-controlled element and its time is the same each cycle (N.A. means performance rating is not applicable). Element *c* is performed while the machine is performing its cycle (element *d*). Element *e* is a worker-controlled irregular element performed every six cycles. The machine is idle during elements *a*, *b*, and *e*. Four product units are produced each cycle. The machine allowance is zero, and a 15% PFD allowance is applied to the manual portion of the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 140% on all manual elements for 7 actual hours worked on an 8-hour shift, determine (c) how many units will be produced and (d) what the worker's efficiency will be.

Work Element	*a*	*b*	*c*	*d*	*e*
Observed time (min)	0.65	0.50	0.50	0.55	1.14
Performance rating (%)	90	100	120	N.A.	80

17 The continuous timing method was used to obtain the times for a worker-machine task. Only one cycle was timed. The observed time data are recorded in the table below. Elements *a*, *b*, *c*, and *e* are worker-controlled elements. Element *d* is machine controlled. Elements *a*, *b*, and *e* are external to the machine-controlled element, while element *c* is internal. There are no irregular elements. All worker-controlled elements were performance rated at 80%. The PFD allowance is 15% and the machine allowance is 20%. Determine (a) the normal time and (b) standard time for the cycle. (c) If worker efficiency is 100%, how many units will be produced in one 9-hour shift? (d) If the actual time worked during the shift is 7.56 hours, and the worker performance is 120%, how many units would be produced?

Worker element (min)	*a* (0.65)	*b* (1.80)	*c* (4.25)	*e* (5.45)
Machine element (min)			*d* (4.00)	

18 The continuous stopwatch timing method was used to obtain the observed times for a worker-machine task. Only one cycle was timed. The data are recorded in the table below. The times listed indicate the stopwatch reading at the end of the element. Elements *a*, *b*, and *d* are worker-controlled elements. Element *c* is machine controlled. Elements *a* and *d* are external to the machine-controlled element, while element *b* is internal. Every four

cycles, there is an irregular worker element that takes 1.32 min rated at 100% performance. For determining the standard time, the PFD allowance is 15% and the machine allowance is 30%. Determine (a) the normal time and (b) standard time for the cycle. (c) If worker efficiency is 100%, how many units will be produced in one 8-hour shift? (d) If the actual time worked during the shift is 6.86 hours, and the worker performance is 125%, how many units will be produced?

Worker Element	Description of Worker Element	Time (min)	Performance Rating (%)	Machine Element	Description of Machine Element	Time (min)
a	Acquire workpart from tray, cut to size, and load into machine	1.24	100	(idle)		
b	Enter machine settings for next cycle	4.24	120	*c*	Automatic cycle controlled by machine settings entered in previous cycle	4.54
d	Unload machine and place part on conveyor	5.09	80	(idle)		

19 The snapback timing method in direct time study was used to obtain the times for a worker-machine task. The recorded times are listed in the table below. Element *d* is a machine-controlled element and the time is constant. Elements *a*, *b*, *c*, *e*, and *f* are operator-controlled and were performance rated at 90%. Element *f* is an irregular element, performed every five cycles. The operator-controlled elements are all external to machine-controlled element *d*. The machine allowance is zero, and the PFD allowance is 13%. Determine (a) the normalized time for the cycle and (b) the standard time for the cycle. The worker's actual time spent working during an 8-hour shift was 7.08 hours, and he produced 400 units of output during this time. Determine (c) the worker's performance during the operator-controlled portions of the cycle and (d) the worker's efficiency during this shift.

Element	*a*	*b*	*c*	*d*	*e*	*f*
Observed time (min)	0.14	0.25	0.18	0.45	0.20	0.62

20 For a worker-machine task, the continuous-timing method was used to obtain the times indicated in the table below. Element *c* is a machine-controlled element and the time is constant. Elements *a*, *b*, *d*, and *e* are operator-controlled and were performance rated at 100%; however, element *d* is performed during the machine-controlled element *c*. The machine allowance is 16%, and the PFD allowance is 16%. Determine (a) the normal time and (b) standard time for the cycle. (c) If the operator works at 140% of standard performance in production and two parts are produced each cycle, how many parts are produced if the total time worked during an 8-hour day is 7.4 hours? (d) For the number of parts computed in (c), what is the worker's efficiency for this shift?

Element	*a*	*b*	*c*	*d*	*e*
Observed time (min)	0.30	0.65	1.65	1.90	2.50

21 For a certain repetitive task, the snapback timing method was used to obtain the average work element times and performance ratings listed in the table below. Elements *a*, *b*, and *c* are worker-controlled. Element *d* is a machine-controlled element and its time is the same each cycle (N.A. means performance rating is not applicable). Element *c* is performed while the machine is performing its cycle (element *d*). Element *e* is a worker-controlled irregular element performed every six cycles. The machine is idle during elements *a*, *b*, and *e*. One part is produced each cycle. The machine allowance is 15%, and a 15% PFD allowance is applied to the manual portion of the cycle. Determine (a) the normal time and (b) the standard time for this cycle. If the worker's performance during actual production is 130% on all manual elements for 7.3 actual hours worked on an 8-hour shift, (c) how many units will be produced and (d) what is the worker's efficiency?

Work Element	*a*	*b*	*c*	*d*	*e*
Observed time (min)	0.47	0.58	0.70	0.75	2.10
Performance rating (%)	90	80	110	N.A.	85

22 The work element times for a repetitive work cycle are listed in the table below, as determined in a direct time study using the snapback timing method. Elements *a* and *b* are operator-controlled. Element *c* is a machine-controlled element and its time is constant. Element *d* is a worker-controlled irregular element performed every 10 cycles. Elements *a* and *b* were performance rated at 90%, and element *d* was performance rated at 75%. The worker is idle during element *c*, and the machine is idle during elements *a*, *b*, and *d*. One product unit is produced each cycle. No special allowance is added to the machine cycle time (element *c*), but a 15% allowance factor is applied to the total cycle time. (a) Determine the standard time for this cycle. If the worker produced 190 units on an 8-hour shift during which 7 hours are actually worked, (b) what was the worker's efficiency, and (c) what was his performance during the operator-paced portion of the cycle?

Work element	*a*	*b*	*c*	*d*
Observed time (min)	0.75	0.30	1.62	1.05

Number of Cycles

23 Seven cycles have been observed during a direct time study. The mean for the largest element time is 0.85 min, and the corresponding sample standard deviation *s* is 0.15 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10% of the true mean, how many more observations should be taken?

24 Nine cycles have been observed during a time study. The mean for the largest element time is 0.80 min, and the corresponding sample standard deviation *s* is 0.15 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10 min of the true mean, how many more observations should be taken?

25 Six cycles have been observed in a direct time study. The mean for the largest element time is 0.82 min, and the corresponding sample standard deviation *s* is 0.11 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within ±0.10 min of the true mean, how many more observations should be taken?

26 Ten cycles have been observed during a direct time study. The mean time for the longest element was 0.65 min, and the standard deviation calculated on the same data was

0.10 min. If the analyst wants to be 95% confident that the mean of the sample was within ±8% of the true mean, how many more observations should be taken?

27 Six cycles have been observed during direct time study. The mean time for the longest element was 0.82 min, and the standard deviation calculated on the same data was 0.13 min. If the analyst wants to be 90% confident that the mean of the sample was within ±0.06 min of the true mean, how many more observations should be taken?

28 Six cycles have been observed during a direct time study. The mean for the largest element time is 1.00 min, and the corresponding sample standard deviation *s* is 0.10 min. (a) Based on these data, what is the 90% confidence interval on the 1.0 min element time? (b) If the analyst wants to be 90% confident that the mean of the sample was within ±10% of the true mean, how many more observations should be taken?

29 A total of 9 cycles has been observed during a direct time study. The mean for the largest element time is 1.30 min, and the corresponding sample standard deviation *s* is 0.20 min. (a) Based on these data, what is the 95% confidence interval on the 1.30 min element time? (b) If the analyst wants to be 98% confident that the mean of the sample was within ±5% of the true mean, how many more observations should be taken?

Performance Rating

30 One of the traditional definitions of standard performance is a person walking at 3.0 miles per hour. Given this, what is the performance rating of a long-distance runner who breaks the four-minute mile?

31 In 1982, the winner of the Boston Marathon was Alberto Salazar, whose time was 2 hr, 23 min and 3.2 sec. The marathon race covers 26 miles and 385 yards. Given that one of the traditional definitions of standard performance is a person walking at 3.0 miles per hour, what was Salazar's performance rating in the race?

Predetermined Motion Time Systems

1 Overview of Predetermined Motion Time Systems
  1.1 The PMTS Procedure
  1.2 PMTS Levels and Generations
2 Methods-Time Measurement
  2.1 MTM-1
  2.2 Other MTM Systems
3 Maynard Operation Sequence Technique
  3.1 Basic MOST
  3.2 Additional Versions of MOST
  3.3 MOST for Windows

The controversial step in direct time study is performance rating. It is based on a judgment by the time study analyst, and there is always room for disagreement between the worker doing the task and the analyst evaluating the worker's pace. An alternative to direct time study that does not include the performance-rating step, at least not directly, is the family of work measurement techniques known as predetermined motion time systems.

A ***predetermined motion time system*** (PMTS), also called a ***predetermined time system***, is a database of basic motion elements and their associated normal time values, together with a set of procedures for applying the data to analyze manual tasks and establish standard times for the tasks.[1] The PMTS database is most readily conceptualized as a set of tables listing time values that correspond to the basic motion elements, the lowest level in our hierarchy of manual work activity. They include motions such as reach, grasp, move, and release. The time required to perform these basic motions usually depends on certain work variables. For example, the time to reach for an

[1]The Work Measurement and Methods Standards Committee (ANSI Standard Z94.11-1989) defines a predetermined motion time system as follows: "An organized body of information, procedures, techniques, and motion times employed in the study and evaluation of manual work elements. The system is expressed in terms of the motions used, their general and specific nature, the conditions under which they occur, and their previously determined performance times."

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object increases with the distance of the reach, among other factors. The time to move an object depends on its weight and the distance it is moved. The normal time values in the database have been determined based on extensive research of manual work activity, usually consisting of frame-by-frame analysis of motion pictures of the activity.

In this chapter we provide a general discussion of predetermined motion time systems, which were first identified around 1925 (Historical Note 1). We then examine two important examples of these systems: methods-time measurement (MTM) and the Maynard Operation Sequence Technique (MOST).

HISTORICAL NOTE 1 PREDETERMINED MOTION TIME SYSTEMS

The notion that manual work consists of basic motion elements is attributed to Frank B. Gilbreth, who was a pioneer in the subject of motion study. His studies resulted in the first methodical classification of motion elements, which he called ***therbligs*** (Gilbreth spelled backwards—well, almost). The 17 therbligs consist mostly of arm and hand motions.

Asa B. Segur is credited with developing the first commercial predetermined motion time system, called Motion-Time Analysis (MTA). Segur was aware of Gilbreth's classification scheme for basic motion elements, and MTA's 17 basic motion elements agree closely with Gilbreth's 17 therbligs. Segur began the development of MTA in 1922 by analyzing motion picture films taken during World War I of operators performing factory tasks. The films were originally intended for use as training materials for blind and otherwise handicapped workers, but Segur constructed his system of motions and times from them. Supposedly for commercial reasons, he limited public access to MTA, and this is one of the reasons why his system is no longer used. However, his work influenced those who followed him in developing other predetermined motion time systems.

One of these analysts was Joseph H. Quick, who developed the Work-Factor system between 1934 and 1938. The system was based on the analysis of large numbers of motion picture films, snapshots, stroboscopic lighting techniques, and stopwatch studies taken of many different kinds of industrial operations. Among Quick's contributions in PMTS research were his studies involving cognitive work, such as visual inspection and similar technical labor.

Harold B. Maynard stands as one of the more important figures in the development of predetermined motion time systems. He is largely responsible for the Methods-Time Measurement (MTM) system, and the Pittsburgh-based consulting firm of H. B. Maynard and Company bears his name. In 1946, Maynard began development of MTM with colleagues G. J. Stegemerten and J. L. Schwab under the sponsorship of the Westinghouse Electric Company. Their study began by analyzing movies of drilling operations and relating the work patterns to therbligs. The therblig classification was found to be wanting in some respects, and many of the basic motion elements were revised, renamed, or removed. MTM became the most successful and widely used first level PMTS after its release in 1948. The MTM database of motion elements was used

in the development of many higher-level systems, including those in the MTM family (see Table 5 later in this chapter).

Other individuals who should be mentioned for their efforts in developing PMT systems include G. B. Bailey and R. Presgrave, who developed Basic Motion Timestudy in 1950; R. M. Crossan and H. W. Nance, who developed Master Standard Data in the late 1950s; and G. C. Heyde, who developed the Modular Arrangement of Predetermined Time Standards (MODAPT) in the mid 1960s.

Kjell B. Zandin is largely responsible for the development of the Maynard Operation Sequence Technique (MOST). Starting in the late 1960s, he led a group at the Swedish Division of H. B. Maynard & Company in studying applications of MTM. It was observed that there were similarities in the motion patterns and MTM sequences of many manual operations. This ultimately resulted in the definition of three principal motion groups, which were called sequence models in the MOST system. MOST was first introduced in Sweden in 1972 and in the United States in 1974. Today, it is one of the most widely used PMT systems.

Subsequent further development of predetermined motion time systems has been in the computerization of several of these systems as commercial products. Examples include MOST for Windows and MTM-LINK.

1 OVERVIEW OF PREDETERMINED MOTION TIME SYSTEMS

This section provides a general introduction to PMT systems. First, we discuss the usual procedure in applying a predetermined motion time system to a task. One of the significant advantages of a PMTS is that it can be used to determine a standard time for a task before that task is in production. It can also be used to set a standard for an existing task. We then indicate the differences among PMT systems and how they can be classified.

1.1 The PMTS Procedure

All PMT systems are applied using the same general procedure. The differences are in the details. To apply a predetermined motion time system to a task, the analyst must accomplish the following steps:

1. Synthesize the method that would be used by a worker to perform the task (or analyze the method that is being used by a worker in an existing task). The method is described in terms of the basic motions comprising the task, based on a defined workplace layout and set of tools (if tools are used).[2]
2. Retrieve the normal time value for each motion element, based on the work variables and conditions under which the element will be (is) performed. Sum the normal times for all motion elements to determine the normal time for the task.

[2]Different predetermined motion time systems have different basic motion definitions, and the analyst must describe the method using the definitions and conventions of the particular system.

3. Evaluate the method to make improvements by eliminating motions, reducing distances, introducing special tools, using simultaneous right and left hand motions, and so on. The objective of this evaluation is to reduce the normal time. The evaluation process is facilitated by the detailed listing of basic motion elements and corresponding time values.
4. Apply allowances to the normal time to determine the standard time for the task.

If the emphasis in the application is methods improvement, then step 4 may be omitted. If the emphasis is setting a time standard, then step 4 is required.

The basic time values in PMT systems do not include any allowances, so the user organization must add an allowance for personal time, fatigue, and delays according to its policies on allowances. This is step 4 in our PMTS procedure outlined above. The following equation is the usual way in which allowances are applied to determine the time standard for a task:

$$T_{std} = T_n(1 + A_{pfd}) \tag{1}$$

where T_{std} = standard time, min; T_n = normal time, which is the sum of the basic motion time values that make up the task (step 2 with possible adjustments resulting from step 3 in the PMTS procedure), min; and A_{pfd} = PFD allowance factor.

A performance rating step is not necessary in a predetermined motion time system, because the basic motion times are already rated at standard (100%) performance. However, different PMT systems use different definitions of standard performance. MTM and MOST use the definition of "daywork" standard performance. If the organization using either of these systems defines standard performance differently, then a universal adjustment must be made to all time values in the database to bring these data into agreement with the organization's definition.

Similarly, for predetermined motion time systems other than MTM or MOST, an adjustment in all data values may be necessary to achieve agreement with the organization's definition of standard performance. For example, Motion Time Analysis (MTA) time values were determined based on a faster working pace, and therefore the basic motion times are less than MTM times by about 20%. Therefore, if a company were to use the MTA system but wanted to set standards using the "daywork" as the standard performance level, then all of the MTA times must be multiplied by 125%.

1.2 PMTS Levels and Generations

There is a variety of predetermined motion time systems. Although we might think that the basic motions of the human body should be the same in all PMT systems, some PMT systems combine basic motions into larger motion elements. Thus, we have various levels of PMT systems. First-level systems use the basic motion elements (e.g., reach, grasp), while higher-level systems combine several basic motion elements into ***motion aggregates***, combined motion sequences that are commonly used in work situations. For example, the basic motions ***reach*** and ***grasp*** might be combined into one element called

get for use in a higher-level PMT system. This motion aggregate would be common in assembly work, for example.

First-level PMT systems tend to be very detailed, with body motions differentiated very precisely in their databases. For example, ***reach*** and ***grasp*** are typically distinguished as two separate motion elements, even though they are commonly carried out as one smooth arm and hand movement. The times for the basic motions tend to be very short, sometimes much less than a second. Higher-level systems use condensed databases, with fewer body motions contained in the tables and longer time values for each motion sequence. The tables are simplified, with fewer work variables that must be identified about the task by the analyst.

Chronologically, first-level PMT systems were the first to be developed, and then second- and higher-level systems were subsequently constructed based on the first-level systems. Because of this chronological development of the systems, the level of the system usually corresponds to the ***generation*** of the system. First-level PMT systems are called first generation systems, and the subsequent systems are second and third generations. For example, MTM-1 is first generation. MTM-2 is second generation and is based on MTM-1. MTM-3 is a third generation MTM system.

Methods descriptions in first-level systems consist of long lists of elements, and the time to determine a standard for the task is long (high application speed ratio). For example, using MTM-1, a first-level system, the time to set a standard is a factor of about 250 compared to the duration of the task time [7]. For a task of 1 minute, it takes over 4 hours of analyst time to establish the standard time. Higher-level PMT systems require less application time due to their use of elements that combine basic motions. For example, MTM-2 is a higher-level system that is based on MTM-1, and the time to set a standard is a factor of 100 compared to the task time. The cost of this greater convenience is that the accuracy of higher-level systems tends to be less than for a first-level system. The general characteristics of PMT system levels are summarized in Table 1.

More than 50 predetermined motion time systems have been developed over the years. Most of these systems are no longer used. Table 2 provides a brief description of the major PMT systems that were developed, with an emphasis on those that are commercially important today. The most widely used systems today are based on MTM. Section 2 discusses MTM, both the first-level system (MTM-1) and some of its higher-level versions. Section 3 discusses MOST, a widely used higher-level PMTS based on MTM.

TABLE 1 Characteristics of PMT System Levels

Characteristic	First-Level PMTS	Higher-Level PMTS
Accuracy	Most accurate	Less accurate
Application time	Much time to set standard	Less time to set standard
Suited to specific types of tasks	Highly repetitive	Repetitive or batch
Cycle times	Short cycle (e.g., 1 min)	Longer cycle times feasible
Motion elements	Basic motions	Aggregates of basic motions
Methods description	Very detailed	Less detailed, easier to apply
Flexibility of application	Highest flexibility	Less flexibility

TABLE 2 Summary of Predetermined Motion Time Systems

Motion-Time Analysis (MTA): A first-level system developed by A. B. Segur in the mid- and late 1920s. Its primary importance today stems from the fact that it was the first PMTS to be developed and applied. Today it is seldom used if at all. There are 17 basic motion elements in MTA, involving one or more body movements. The elements are classified as positive (considered useful to the task), negative (motions that should be reduced or eliminated), or either (might be positive or negative).

Work-Factor (WF): A first-level PMTS developed by J. H. Quick, W. J. Shea, and R. E. Koehler in the mid-1930s, with applications starting in 1938. It is still used today and at least one simplified version has been developed to reduce application time. Coverage includes both factory and office work.

Methods-Time Measurement (MTM): Developed by H. B. Maynard, G. J. Stegemerten, and J. L. Schwab in the mid-1940s. The original MTM was a generic, first-level system published as a book in 1948. Today, that original PMTS is known as MTM-1, and several higher-level and specific PMT systems have been developed based on MTM-1. We describe the MTM family in Section 2.

Basic Motion Timestudy (BMT): Developed by R. Presgrave, G. B. Bailey, and J. A. Lowden in the early 1950s. Its initial applications were in 1952, and it was published in book form in 1958. It is considered a first-level system, but some of its motion elements are combinations of therbligs. At least one second-level PMTS has been developed based on BMT.

Master Standard Data (MSD): One of the first second-level systems. It is based on MTM-1 and was developed in Canada by R. M. Crossan and H. W. Nance in the late 1950s (published as a book in 1962) to reduce the analyst's time needed to set a time standard for manual production operations with fewer than 100,000 cycles per year. Its coverage emphasizes reach-grasp-move-release motions that are common in such operations.

Maynard Operation Sequence Technique (MOST): Developed by K. B. Zandin and others at the Swedish Division of H. B. Maynard Company as a higher-level PTMS based on MTM-1 and MTM-2. Its coverage emphasizes production operations and material handling. We describe MOST in Section 3.

Modular Arrangement of Predetermined Time Standards (MODAPTS): A second-level PTMS developed by C. Heyde and others in Australia during the mid-1960s. It is based on MSD, MTM-1, and MTM-2. The basic motion element in MODAPTS is a finger movement, and other body motions are expressed in terms of this element. Coverage includes production operations and material handling, with a special version covering office work.

Source: Compiled from [1], [7], and other sources.

2 METHODS-TIME MEASUREMENT

Methods-Time Measurement (MTM) is a PMTS that can be used to analyze the method for performing a given task and to set a time standard for the task. MTM is defined by its developers as follows [4]:

> Methods-Time Measurement is a procedure which analyzes any manual operation or method into the basic motions required to perform it and assigns to each motion a predetermined time standard which is determined by the nature of the motion and the conditions under which it is made.

The hyphenation of "Methods-Time" was intended by its developers to emphasize the important connection between the basic motions used to perform a task and the time values associated with these motions. MTM is an analysis tool by which the method for a task is divided into its component elements to determine if any methods improvements can be made (step 3 in the PMTS procedure). Only after this analysis should a standard time be established.

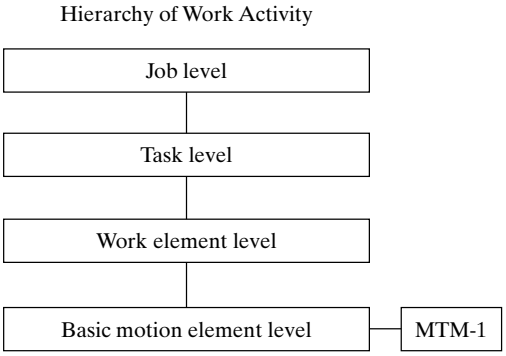


Figure 1 The position of MTM motion elements in our work hierarchy.

MTM is actually a family of predetermined motion time systems. The original MTM is now called MTM-1. It is the first-level PMTS upon which all of the other MTM systems are based. We describe MTM-1 in Section 2.1. Other versions of MTM are briefly discussed in Section 2.2. Training in the members of the MTM PMTS family is available through the MTM Association, located in Des Plaines, Illinois.[3]

2.1 MTM-1

In our hierarchy of work activity, MTM-1 operates at the basic motion element level, as illustrated in Figure 1. Most of the MTM-1 basic motions involve hand and arm movements, although elements are also provided for eye, leg, foot, and body actions. Many of the basic motions correspond to the original therbligs as developed by Frank Gilbreth. However, the definitions of these basic motions have been adjusted to conform to the needs of MTM, and new elements not included among the original therbligs have been added.

Time units in MTM are called TMUs (time measurement units). MTM was developed by studying motion pictures of work activity, and the time units for MTM were originally defined as the time per frame of motion picture film. This proved awkward, and the TMU is now defined as follows:

$$1 \text{ TMU} = 0.00001 \text{ hr} = 0.0006 \text{ min} = 0.036 \text{ sec}$$

Conversely, there are 100,000 TMUs in 1 hr, 1667 TMUs in 1 min, and 27.8 TMUs in 1 sec.

Table 3 defines the MTM-1 motion elements, and Table 4 presents a tabulation of their time values. As with other PMT systems, the analyst must describe the task in terms of these basic motion elements, taking note of the work variables that influence the element time and accounting for simultaneous right-hand and left-hand motions,

[3]MTM Association, 1111 East Touhy Avenue, Des Plaines, IL 60018; their Web site can be accessed at www.mtm.org.

TABLE 3 Summary of MTM-1 Motion Elements

Motion Element (Symbol) and Description

Reach (R): A basic motion element involving movement of the hand or fingers. Its purpose is to move the hand or finger to a new destination. The corresponding therblig is "transport empty." If the hand is holding something during the motion, it is still classified as a reach if its primary purpose is to reposition the hand or fingers and not to move the object—for example, reaching for an object while holding a cigarette. Reach depends on two work variables: (1) the distance moved and (2) the case under which the reach is performed. The case refers to such factors as whether the reach is toward an object in a known location or the object is jumbled amongst other objects and must be searched out. The five cases are identified in Table 4 (a), which provides time values as a function of distance and case. If the hand is in motion before and/or after the reach, then the time is reduced slightly, as indicated in the table under "hand in motion" for cases A and B. The complete symbol for a given reach element in MTM-1 includes the distance and the case (e.g., R10C).

Grasp (G): A motion element used by the fingers and hand to gain control of one or more objects. It is commonly employed as a prerequisite motion for performing the next basic motion, which is likely to be a move—for example, grasping an object prior to moving it. There are five categories of grasp: (1) pickup, (2) regrasp, (3) transfer, (4) select, and (5) contact. Within the pickup and select categories, there are several cases that must be distinguished. Table 4 (b) defines the various categories and cases and lists the times for them. The complete symbol for a grasp includes category and case (e.g., G1C3).

Move (M): A hand or finger motion whose primary purpose is to relocate an object. The corresponding therblig is "transport loaded." The move element includes pushing or sliding an object across a surface, so long as the hand controls the object. Move depends on three work variables: (1) the weight of the object being moved, (2) the distance of the move, and (3) the case under which the move is performed. There are three cases, as described in Table 4 (c). The weight variable is divided into classes, and the user selects the next higher weight given in the table for the object weight in the given application. The complete symbol for move includes the distance, case, and weight—for example, M6B10 symbolizes a move of 6 in. under case B for an object weight of 10 lb. The following formula must be used to determine the TMU value from Table 4 (c):

$$\text{TMU} = \text{Constant} + \text{Factor} \times (\text{TMU value from table})$$

where Constant is the constant value from the table for the object weight in the application, TMU; Factor is the factor value from the table for the object weight; and (TMU value from table) is the TMU value from the table for the corresponding distance of the move and the move case. For example, given the symbol M6B10, the formula would be used as follows:

$$\text{TMU} = 3.9 + 1.11(8.9) = 13.8 \text{ TMU (about } {}^{1}\!/_{2} \text{ sec)}$$

Position (P): A relatively short hand motion (no greater than 1 in.) employed to align, orient, or engage one object relative to another. It is usually preceded by a move. ***Align*** refers to the relative positioning of longitudinal axes of the objects; ***orient*** refers to the relative positioning of rotational axes of the objects; and ***engage*** refers to the insertion of one object into the other following either an alignment or an orientation. Accordingly, the position element may occur several times together depending on the number of separate positioning moves required. The work variables that affect the time for a position motion are the following: (1) pressure required to achieve the fit, (2) symmetry of the objects, and (3) ease of handling. Pressure to achieve fit is divided into three categories: loose (class 1), close (class 2), and exact (class 3). Symmetry is also divided into three categories: symmetrical (S)—for example, a round peg in a round hole; nonsymmetrical (NS)—for example, a key inserted into a lock; and semisymmetrical (SS), which is any case between S and NS. Ease of handling is simply divided into easy (E) and difficult (D). The categories are listed in Table 4 (d) along with the time to perform the element. The complete symbol for a position includes the three work variables (e.g., P3NSD).

Release (RL): A hand and finger motion element whose purpose is to surrender control of an object. There are only two cases of release: (1) the fingers open to effect the release and (2) contact release with no finger motion. The complete symbol for release identifies the case (e.g., RL1 or RL2). Times are given in Table 4 (e).

Disengage (D): A hand and finger motion that causes the separation of two objects from one another, where the objects were previously held together by some force. Thus, disengagement breaks the force, resulting in a recoil action by the hand. The work variables that affect the time of a disengage motion are (1) class of fit and (2) ease of handling. Classes

(*continued*)

TABLE 3 (*continued*)

Motion Element (Symbol) and Description
of fit are loose (slight effort required, class 1), close (normal effort, class 2), and tight (considerable effort, class 3). Ease of handling is either easy (E) or difficult (D). The disengage symbol indicates the two variables (e.g., D3E). Times are given in Table 4 (f).
Turn (T): A basic motion element that involves rotation of the hand and wrist about the long axis of the forearm. The hand can either be holding an object (e.g., turning a dial on a machine) or empty. Turn depends on two work variables: (1) the degrees of turn and (2) resistance to the turn. Table 4 (g) lists the time values as a function of these factors. The maximum turn in the table is 180°, which is imposed by the anatomical structure of the elbow and forearm. Resistance to turn is indicated by force of resistance, divided into three categories in the table. The complete symbol for turn includes both variables; for example, T90L symbolizes a turn of 90° and a large resisting force.
Apply pressure (AP): An MTM-1 element that involves the application of force, but the force results in little or no movement. The force is usually applied by the hand and/or fingers, but the element applies to a force applied by any body member. There are two categories of apply pressure: (1) apply pressure alone (APA) and (2) apply pressure preceded by a regrasp (APB). Time values for these two cases are given in Table 4 (h).
Eye travel (ET), **eye focus (EF)**, and **reading**: Basic eye usage motions that are important enablers for performing manual work. However, the effect of the eyes during a motion element is usually accounted for in the tabulated TMU values. For example, a reach motion that requires greater hand-eye coordination takes longer; for example, compare Case C and Case A in Table 4 (a). On the other hand, there are situations when eye travel and/or eye focus are prerequisites for subsequent motions. In these cases, they must be counted as separate basic elements. Times for the two basic eye usage motions are given in Table 4 (i). In addition, MTM-1 uses a normal time for reading text as 5.05 TMUs per word. Thus, reading a 200 word paragraph would be allowed 1010 TMUs (about 36 sec).
Body, leg, and foot motions: Additional basic motions that involve moving the body, one or both legs, and/or one or both feet. The motions include walking, bending, stooping, standing from a seated position, and sitting from a standing position. These activities are distinct from those performed at a workbench, most likely with the worker in a sitting position. These elements and others are identified in Table 4 (j), together with their symbols and normal times.
Simultaneous motions: Basic motion elements that are performed at the same time. In general, it is desirable for more than one body member to be moving simultaneously. For example, it is desirable for both the right hand and left hand to be active during the task, not only sequentially but also simultaneously. When basic motion elements can be combined simultaneously, the time required for the combination is the greater of the two parallel motions. This is called the "limiting principle," which was first proposed by the original developers of MTM (Maynard, Stegemerten, and Schwab, Historical Note 1). Certain combinations of motions are more difficult to perform simultaneously than others. The degrees of difficulty can be classified as follows: (1) easy to perform simultaneously, (2) can be performed simultaneously with practice, and (3) difficult to perform simultaneously. Table 4 (k) indicates the degree of difficulty for various combinations of basic motion elements. In degrees of difficulty (1) and (2), the limiting principle applies, assuming sufficient practice is allowed for the worker in (2). In degree of difficulty (3), both times should be added.

internal elements when the task involves operation of a machine, and other special circumstances. The analyst then retrieves the time for each basic motion element and sums the times to obtain the normal time for the task. Since the MTM time values do not include any allowances, the analyst must add the proper allowance according to company policy for the type of work being measured. Having stated this, we are obligated to mention that MTM-1 should not be used in industrial work measurement applications until the analyst has received proper training in the technique.

2.2 Other MTM Systems

Additional members of the MTM family have been introduced to satisfy various user needs. Several functional systems have been developed for specific work situations such as clerical activity, machine shop work, and electronic testing. Also, second- and third-level

TABLE 4 (a) Normal Time Values for MTM-1 Motion Element: **Reach** (R)

Distance		Time in TMU				Hand in Motion		Case and Description
cm	inches	A	B	C or D	E	A	B	
< 2.0	< 0.75	2.0	2.0	2.0	2.0	1.6	1.6	**A** Reach to object in fixed location, or to object in other hand or on which other hand rests.
2.5	1	2.5	2.5	3.6	2.4	2.3	2.3	
5.1	2	4.0	4.0	5.9	3.8	3.5	2.7	
7.6	3	5.3	5.3	7.3	5.3	4.5	3.6	**B** Reach to single object in location that may vary slightly from cycle to cycle.
10.1	4	6.1	6.4	8.4	6.8	4.9	4.3	
12.5	5	6.5	7.8	9.4	7.4	5.3	5.0	
15.2	6	7.0	8.6	10.1	8.0	5.7	5.7	
17.8	7	7.4	9.3	10.8	8.7	6.1	6.5	**C** Reach to object jumbled with other objects in a group so that search and select occur.
20.3	8	7.9	10.1	11.5	9.3	6.5	7.2	
22.9	9	8.3	10.8	12.2	9.9	6.9	7.9	
25.4	10	8.7	11.5	12.9	10.5	7.3	8.6	
30.5	12	9.6	12.9	14.2	11.8	8.1	10.1	**D** Reach to a very small object or where accurate grasp is required.
35.6	14	10.5	14.4	15.6	13.0	8.9	11.5	
40.6	16	11.4	15.8	17.0	14.2	9.7	12.9	
45.7	18	12.3	17.2	18.4	15.5	10.5	14.4	
50.8	20	13.1	18.6	19.8	16.7	11.3	15.8	**E** Reach to indefinite location to get hand in position for body balance or next motion or out the way.
55.9	22	14.0	20.1	21.2	18.0	12.1	17.3	
61.0	24	14.9	21.5	22.5	19.2	12.9	18.8	
66.0	26	15.8	22.9	23.9	20.4	13.7	20.2	
71.1	28	16.7	24.4	25.3	21.7	14.5	21.7	
76.2	30	17.5	25.8	26.7	22.9	15.3	23.2	
Additional		0.4	0.7	0.7	0.6	TMU per 2.54 cm > 76 cm (per 1.0 in > 30 in.)		

TABLE 4 (b) Normal Time Values for MTM-1 Motion Element: **Grasp** (G)

Type of Grasp	Case	Time, TMU	Description and Object Dimensions	
Pickup	1A	2.0	Any size object, by itself	
	1B	3.5	Object very small or lying close against a flat surface	
	1C1	7.3	Interference with grasp on bottom and one side of cylindrical object	Diameter > 1.3 cm (0.5 in.)
	1C2	8.7		Diameter 0.6 to 1.3 cm (0.25 to 0.5 in.)
	1C3	10.8		Diameter < 0.6 cm (0.25 in.)
Regrasp	2	5.6	Change grasp without relinquishing control	
Transfer	3	5.6	Control transferred from one hand to other	
Select	4A	7.3	Object jumbled with other objects so that search and select occur	Size larger than 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.)
	4B	9.1		0.6 × .6 × .3 cm (.25 × .25 × .12 in) to 2.5 × 2.5 × 2.5 cm (1 × 1 × 1 in.)
	4C	12.9		Size smaller than .6 × .6 × .3 cm (.25 × .25 × .12 in.)
Contact	5	0	Contact, sliding, or hook grasp	

TABLE 4 (c) Normal Time Values for MTM-1 Motion Element: **Move** (M)

Distance		Time in TMU			Hand in motion	Weight up to	Formula Parameters		Case and Description
cm	inches	A	B	C	B	kg (lb)	Constant	Factor	
< 2.0	< 0.75	2.0	2.0	2.0	1.7				**A** Move object to other hand or against stop.
2.5	1	2.5	2.9	3.4	2.3	1.1 (2.5)	0	1.00	
5.1	2	3.6	4.6	5.2	2.9				
7.6	3	4.9	5.7	6.7	3.6	3.4 (7.5)	2.2	1.06	
10.1	4	6.1	6.9	8.0	4.3				**B** Move object to approximate or indefinite location.
12.5	5	7.3	8.0	9.2	5.0	5.7 (12.5)	3.9	1.11	
15.2	6	8.1	8.9	10.3	5.7				
17.8	7	8.9	9.7	11.1	6.5	7.9 (17.5)	5.6	1.17	
20.3	8	9.7	10.6	11.8	7.2				
22.9	9	10.5	11.5	12.7	7.9	10.2 (22.5)	7.4	1.22	**C** Move object to exact location.
25.4	10	11.3	12.2	13.5	8.6				
30.5	12	12.9	13.4	15.2	10.0	12.5 (27.5)	9.1	1.28	
35.6	14	14.4	14.6	16.9	11.4				
40.6	16	16.0	15.8	18.7	12.8	14.7 (32.5)	10.8	1.33	
45.7	18	17.6	17.0	20.4	14.2				
50.8	20	19.2	18.2	22.1	15.6	17.0 (37.5)	12.5	1.39	
55.9	22	20.8	19.4	23.8	17.0				
61.0	24	22.4	20.6	25.5	18.4	19.3 (42.5)	14.3	1.44	
66.0	26	24.0	21.8	27.3	19.8				
71.1	28	25.5	23.1	29.0	21.2	21.5 (47.5)	16.0	1.50	
76.2	30	27.1	24.3	30.7	22.7				
Additional		0.8	0.6	0.85	TMU per 2.54 cm > 76 cm (per 1.0 in. > 30 in.)				

TABLE 4 (d) Normal Time Values for MTM-1 Motion Element: **Position** (P)

			Time in TMU	
Class	Description of Fit	Symmetry	Easy to Handle	Difficult to Handle
1	Loose (no pressure required)	S	5.6	11.2
		SS	9.1	14.7
		NS	10.4	16.0
2	Close (light pressure required)	S	16.2	21.8
		SS	19.7	25.3
		NS	21.0	26.6
3	Exact (heavy pressure required)	S	43.0	48.6
		SS	46.5	52.1
		NS	47.8	53.4
Key: S = symmetrical, SS = semi-symmetrical, NS = nonsymmetrical.				

TABLE 4 (e) Normal Time Values for MTM-1 Motion Element: **Release** (RL)

Case	Time in TMU	Description
1	2.0	Normal release performed by opening fingers as an independent motion
2	0	Contact release with no finger motion

TABLE 4 (f) Normal Time Values for MTM-1 Motion Element: **Disengage** (D)

			Time in TMU	
Class	Description of Fit	Height of Recoil	Easy to Handle	Difficult to Handle
1	Loose (very slight effort, blends with subsequent move)	Up to 2.5 cm (1 in)	4.0	5.7
2	Close (normal effort, slight recoil) (1 to 5 in)	2.5 to 12.7 cm	7.5	11.8
3	Tight (considerable effort, hand recoils markedly)	12.7 to 30 cm (5 to 12 in)	22.9	34.7

TABLE 4 (g) Normal Time Values for MTM-1 Motion Element: **Turn** (T)

	Time in TMU for Degrees Turned										
Weight, kg (lb)	30°	45°	60°	75°	90°	105°	120°	135°	150°	165°	180°
Small, up to 0.9 (2)	2.8	3.5	4.1	4.8	5.4	6.1	6.8	7.4	8.1	8.7	9.4
Medium, 1 to 4.5 (2 to 10)	4.4	5.5	6.5	7.5	8.5	9.6	10.6	11.6	12.7	13.7	14.8
Large, 4.5 to 16 (10 to 35)	8.4	10.5	12.3	14.4	16.2	18.3	20.4	22.2	24.3	26.1	28.2

TABLE 4 (h) Normal Time Values for MTM-1 Motion Element: **Apply Pressure** (AP)

Symbol	Time in TMU	Description
APA	10.6	Apply pressure alone
APB	16.2	Apply pressure preceded by regrasp

TABLE 4 (i) Normal Time Values for MTM-1 Motion Element: **Eye Travel** (ET) and **Eye Focus** (EF)

Eye motion	Symbol	Time in TMU	Key to Symbols
Eye travel	ET	$\frac{15.2L}{D}$	L = distance between points from and to which eye travels, D = perpendicular distance from the eye to the line of travel. Maximum time allowed = 20 TMU
Eye focus	EF	7.3	
Reading	(none)	$5.05N$	N = number of words read (330 words/min)

TABLE 4 (j) Normal Time Values for MTM-1 Motion Element: **Body, leg**, and **foot motions** (various symbols given in table)

Motion	Symbol	Time in TMU	Description and Conditions
Sit	SIT	34.7	From standing position
Stand	STD	43.4	From seated position
Turn body	TBC1	18.6	Turn body 45° to 90°, Case 1 – Lagging foot not aligned with leading foot
Turn body	TBC2	37.2	Turn body 45° to 90°, Case 2 – Lagging foot aligned with leading foot
Bend	B	29.0	Bend body forward so hands can reach knees
Stoop	S	29.0	Stoop body forward so hands can reach floor
Arise	AB	31.9	Arise from bent position
Arise	AS	31.9	Arise from stooped position
Kneel	KOK	29.0	Kneel on one knee
Kneel	KBK	69.4	Kneel on both knees
Arise	AKOK	31.9	Arise from kneeling position on one knee
Arise	AKBK	76.7	Arise from kneeling position on both knees
Walk	W*X*FT	5.3 per ft	Walking in ft of distance, X = distance in ft
Walk	W*N*P	15.0/pace	Walking in number of paces, N = number of paces
Walk	W*N*PO	17.0/pace	Walking in number of paces with weight or obstruction, N = number of paces
Leg motion	LM6	7.1	Move leg up to 6 in. any direction
Leg motion	LM*X*	7.1 + 1.2(X-6)	Move leg more than 6 in. any direction, where X = distance of movement
Foot motion	FM	8.5	Foot moves up to 4 in. hinged at ankle
Foot motion	FMP	19.1	Foot moves up to 4 in. hinged at ankle, apply heavy pressure with leg muscles

TABLE 4 (k) MTM-1 Simultaneous Hand and Arm Motion Elements

Motion of One Hand		Reach			Grasp			Move			Position			Disengage	
Motion of other hand	Case or Class	A E	B	C D	1A 2 5	1B 1C	4	A Bm[a]	B	C	1S	1SS 2S	1NS 2SS 2NS	1E 1D	2
Reach	A, E	1	1	1	1	1	1	1	1	2	1	1	2	1	1
	B	1	1	1	1	1	2	1	1	2	2	2	3	1	1
	C, D	1	1	1	1	2	3	2	2	3	3	3	3	2	3
Grasp	1A, 2, 5	1	1	1	1	1	1	1	1	1	1	1	3	1	3
	1B, 1C	1	1	2	1	3	2	1	1	2	3	3	3	3	3
	4	1	2	3	1	2	3	1	2	3	3	3	3	3	3
Move	A	1	1	2	1	1	1	1	1	1	1	1	2	1	1
	B	1	1	2	1	1	2	1	1	1	2	2	3	1	1
	C	2	2	3	1	2	3	1	1	2	3	3	3	2	3
Position	1S	1	2	3	1	3	3	1	2	3	2	3	3	3	3
	1SS,2S	1	2	3	1	3	3	1	2	3	3	3	3	3	3
1NS, 2SS,2NS		2	3	3	3	3	3	2	3	3	3	3	3	3	3
Disengage	1E,1D	1	1	2	1	3	3	1	1	2	3	3	3	1	1
	2	1	1	3	3	3	3	1	1	3	3	3	3	1	1

[a]Bm is Case B with hand in motion.

(*continued*)

TABLE 4 (k) (*continued*)

Key: The cell numbers indicate the degree of difficulty when motions are performed simultaneously.
1 = Easy to perform simultaneously. Use the longest motion element time.
2 = Can be performed simultaneously with practice. Use the longest motion element time.
3 = Difficult to perform simultaneously. Add the times of the two simultaneous motion elements.

Assumptions: All Reach, Grasp, and Move motions are performed within the area of normal vision. In the Position and Disengage motion elements, objects are assumed easy to handle. In general, the degree of difficulty increases if these assumptions are violated.

Motions not included in the table:
Turn: normally degree of difficulty = 1 except when Turn is controlled or with Disengage.
Position Class 3: degree of difficulty = 3.
Disengage Class 3: normally degree of difficulty = 3.
Release: degree of difficulty = 1.

TABLE 5 Members of the MTM Family of Predetermined Motion Time Systems

MTM-1: The first level MTM, in which basic motion elements are used to describe, analyze, and determine the normal time for a manual task. The MTM-1 motion elements and associated times are listed in Table 3 and 4. MTM-1 is best suited to high production operations with relatively short cycle times. The analyst time required to apply MTM-1 is about 250 times the task cycle time.

MTM-2: A second-level MTM system, in which the basic motion elements are combined into motion aggregates in order to reduce the analyst's time to apply the technique. MTM-2 consists of 11 motions and motion aggregates, called *motion categories*. The two most important MTM-2 categories are GET, which combines Reach, Grasp, and Release; and PUT, which combines Move and Position. MTM-2 is suited to operations that are not highly repetitive and the cycle times are greater than 1 min. The analyst time required to apply MTM-2 is about 100 times the task cycle time.

MTM-3: A third-level MTM system designed to further reduce the analyst's time to set a time standard but at some sacrifice in accuracy. MTM-3 has only four motion categories: (1) Handle, (2) Transport, (3) Step and foot motions, and (4) Bend and arise. Tasks that include additional elements, such as eye movements, should not be analyzed using MTM-3. The analyst time required to apply MTM-3 is about 35 times the task cycle time.

MTM-UAS: A third-level MTM system that is suited to applications in batch production. It includes seven basic motion categories: (1) Get and place, (2) Place, (3) Handle tool, (4) Operate, (5) Motion cycles, (6) Body motions, and (7) Visual control. Additional second-level standard data in MTM-UAS cover activities such as fastening, marking, packing, and assembling. The analyst time required to apply MTM-UAS is about 30 times the task cycle time.

MTM-MEK:A third-level system intended for work measurement applications in small lot production with long cycle times and other tasks performed infrequently. Time standards are usually not established in these work situations by conventional time study because the cost of setting the standard is too high. MTM-MEK can be used for these cases and the analyst time is about 5 to 15 times the cycle time.

MTM-HC: A functional PMTS designed for work activities found in the health-care industry. It is described as a standard database by the MTM Association.

MTM-C: A functional MTM system used for work measurement applications of clerical work activity, such as typing or keypunching, filing, reading, and writing. It was developed by an association of banking and other service industries. There are two levels of MTM-C: MTM-C1 and MTM-C2. The difference is that MTM-C1 emphasizes precision and motion detail in its applications, while MTM-C2 emphasizes speed of application. The analyst time to apply MTM-C1 is about 125 times the task cycle time, and about 75 times for MTM-C2.

MTM-V: A functional standard data system developed using MTM-1 for work measurement in machine tool operation. Its work elements include handling of work parts, operating a machine tool and other equipment (e.g., cranes,

(*continued*)

TABLE 5 (*continued*)

fixtures, chucks), and setting up a job for production. The analyst time required to apply MTM-V is about 10 times the task cycle time.

MTM-TE: A functional standard data system designed for work measurement in electronic testing applications. Its work elements cover both manual and mental activities.

MTM-M: A functional MTM system for measuring assembly work that is performed under a stereoscopic microscope.

Source: Estimates of analyst time to apply the various MTM systems are based on [7].

systems are available to reduce the time required to develop time standards. Finally, several computerized systems have been developed based on MTM. Table 5 lists many of these MTM systems with a brief description of each. Readers interested in learning more about the other MTM systems can consult the coverage provided in [6] or the MTM Web site at www.mtm.org.

3 MAYNARD OPERATION SEQUENCE TECHNIQUE

The Maynard Operation Sequence Technique (MOST) is a high-level predetermined motion time system that is based on MTM. It uses the same time units as in MTM: the TMU (time measurement unit), which is 0.00001 hr. MOST is a product of H. B. Maynard and Company, an educational and consulting firm located in Pittsburgh, Pennsylvania.[4] The basic version of MOST, which is now referred to as Basic MOST, was developed at Maynard's Swedish Division around 1967 under the direction of Kjell Zandin (Historical Note 1). Section 3.1, describes the Basic MOST package. Several additional versions of MOST have been developed since Basic MOST was introduced, and these are summarized in Section 3.2. Section 3.3 describes a computerized version of MOST.

3.1 Basic MOST

The focus of Basic MOST is on work activity involving the movement of objects. In fact, the majority of industrial manual work does involve moving objects (e.g., parts, tools) from one location to another in the workplace. The underlying premise of Basic MOST is that the movement of objects consists of a pattern of body motions and actions that is nearly the same for all moves. Only the details of the pattern differ, depending on the object being moved and the circumstances of the work activity. Consistent with this premise, Basic MOST uses motion aggregates (collections of basic motion elements) that are concerned with moving things. The motion aggregates are called ***activity sequence models*** in Basic MOST. In our hierarchy of work activity, an activity sequence model is either at the same level as a work element, or slightly below. Depending on how a work element is defined for a given task, it may include one or more activity sequence models. Rarely would a work element be defined as less than a complete activity sequence model. The relationships are indicated in Figure 2.

[4]MOST is a registered trademark of H. B. Maynard and Company, Inc.

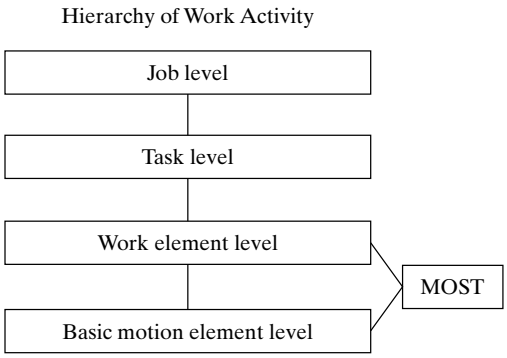


Figure 2 The position of the Basic MOST activity sequence model in our work hierarchy.

There are three activity sequence models in Basic MOST, each of which consists of a standard sequence of actions:

- *General move.* This sequence model is used when an object is moved freely through space from one location to the next (e.g., picking something up from the floor and placing it on a table).
- *Controlled move.* This sequence model is used when an object is moved while it remains in contact with a surface (e.g., sliding the object along the surface) or the object is attached to some other object during its movement (e.g., moving a lever on a machine).
- *Tool use.* This sequence model applies to the use of a hand tool (e.g., a hammer or screwdriver).

The actions in an activity sequence model, called ***sequence model parameters*** in Basic MOST, are similar to basic motion elements in MTM, but there is not a one-to-one correspondence. Some parameters map directly into MTM basic motions, while others are unique to Basic MOST. Let us examine the three sequence models and indicate the standard sequence of model parameters for each.[5]

General Move. The General Move sequence is applicable when an object is moved through the air from one location to another. There are four parameters (actions) in the General Move, symbolized by letters of the alphabet:

A—Action distance, usually horizontal. This parameter is used to describe movements of the fingers, hands, or feet (e.g., walking). The movement can be performed either loaded or unloaded.

B—Body motion, usually vertical. This parameter defines vertical body motions and actions (e.g., sitting, standing up).

[5]Once again, we must caution against the application of MOST without proper training. Guidelines can be obtained from H. B. Maynard & Company in Pittsburgh, Pennsylvania; their Web site can be accessed at www.hbmaynard.com.

G—Gain control. This parameter is used for any manual actions involving the fingers, hands, or feet to gain physical control of one or more objects. It is closely related to the grasp motion element in MTM (e.g., grasp the object).

P—Placement. The placement parameter is used to describe the action involved to lay aside, position, orient, or align an object after it has been moved to the new location (e.g., position the object).

These parameters occur in the following standard sequence in the General Move:

A	B	G	A	B	P	A
Action distance	Body motion	Grasp	Action distance	Body motion	Placement	Action distance

where the first three parameters (A B G) represent basic motions to ***get*** an object, the next three parameters (A B P) represent motions to ***put*** or move the object to a new location, and the final parameter (A) applies to any motions at the end of the sequence, such as ***return*** to original position.

To complete the activity sequence model, each parameter is assigned a numerical value in the form of a subscript or index number that represents the time to accomplish that action. The value of the index number depends on the type of action, its motion content, and the conditions under which it is performed. Table 6 lists the parameters and possible circumstances for the action, together with the corresponding values of the index numbers. When the index values have been entered for all parameters, the time for the sequence model is determined by summing the index values and multiplying by 10 to obtain the total TMUs. The procedure is illustrated in the following example.

Example 1 General Move

Develop the activity sequence model and determine the normal time for the following work activity: A worker walks 5 steps, picks up a small part from the floor, returns to his original position, and places the part on his worktable.

Solution: Referring to Table 6, the indexed activity sequence model for this work activity would be the following:

$$A_{10}B_6G_1A_{10}B_0P_1A_0$$

where A_{10} = walk 5 steps, B_6 = bend and arise, G_1=gain control of small part, A_{10} = walk back to original position, B_0 = no body motion, P_1 = lay aside part on table, and A_0 = no motion. The sum of the index values is 28. Multiplying by 10, we have 280 TMUs (about 10 sec). ■

The General Move is the most common way in which materials are moved and is therefore the most frequently used activity sequence model in Basic MOST. Approximately half of all manual work occurs as General Moves. For assembly work and material handling, the proportion is higher. For machine shop work, the proportion is lower.

TABLE 6 MOST Parameters and Index Values for the General Move Activity Sequence Model

General Move activity sequence model = **A B G A B P A**				
Index	**A** = Action distance	**B** = Body motion	**G** = Gain control	**P** = Placement
0	Close ≤ 5 cm (2 in.)			Hold, Toss
1	Within reach (but > 2 in.)		Grasp light object using one or two hands	Lay aside Loose fit
3	1 or 2 steps	Bend and arise with 50% occurrence	Grasp object that is heavy, or obstructed, or hidden, or interlocked	Adjustments, light pressure, double placement
6	3 or 4 steps	Bend and arise with 100% occurrence		Position with care, or precision, of blind, or obstructed, or heavy pressure
10	5, 6, or 7 steps	Sit or stand		
16	8, 9, or 10 steps	Through door, or Climb on or off, or Stand and bend, or Bend and sit		

Controlled Move. The Controlled Move sequence model is used when an object is moved through a path that is somehow constrained. The object cannot be freely moved through the air. This situation arises when an object is slid across a surface or when the object is attached to something else and it can be moved only through a controlled path. The second case is commonly encountered in the operation of machinery, and the move involves a lever, crank, switch button, or similar entity that is attached to the machinery. The parameters in the Controlled Move include the A, B, and G parameters from the General Move. In addition, three new parameters are introduced:

M—Move, controlled. This parameter is used to describe any manual body motions required to move an object over a controlled path.

X—Process time. Since the Controlled Move can include the operation of machinery, there is often a process time associated with the machinery (e.g., time to turn the workpiece in a lathe). No manual motions are included in this parameter.

I—Align. This parameter is used when manual motions are performed at the end of the controlled move to align objects.

These parameters occur in the following standard sequence in the Controlled Move:

A	B	G	M	X	I	A
Action distance	Body motion	Grasp	Move, controlled	Process time	Align	Action distance

where the first three parameters (A B G) are the basic motions to ***get*** an object, the next three parameters (M X I) represent moving or actuating the object followed by a process time and alignment, and the final parameter (A) provides for a possible ***return*** at the end of the sequence.

TABLE 7 MOST Parameters and Index Values for the Controlled Move Activity Sequence Model

Controlled Move activity sequence model = **A B G M X I A**				
Index	**M** = Move, controlled	**X** = Process time[a]		**I** = Alignment
		Seconds	Minutes	
1	Push, pull, pivot: button, switch, knob (≤ 12 in.)	0.5	0.01	Align to one point
3	Push and pull, turn, open, seat, shift, press: resistance encountered, or high control required, or 2 stages of control (≤ 12 in.); 1 crank of lever.	1.5	0.02	Align to 2 points, Close align (≤ 4 in.)
6	Open and shut, operate, push or pull: with 1 or 2 steps (> 12 in.); 3 cranks of lever.	2.5	0.04	Align to 2 points, Close align (> 4 in.)
10	Manipulate, maneuver, push, or pull with 3, 4, or 5 steps; 6 cranks of lever.	4.5	0.07	Precision align
16	Push or pull with 6, 7, 8, or 9 steps included; 11 cranks of lever.	7.0	0.11	High precision align

[a]For process times longer than those listed in the table, the actual process time in seconds can be multiplied by 2.78 and rounded to the next higher value to obtain the index for the X parameter.

Index values are applied to each parameter as in the General Move. The same A, B, and G index values from Table 6 apply to the Controlled Move. Table 7 lists the index values for the M, X, and I parameters used exclusively for the Controlled Move.

Example 2 Controlled Move

Develop the activity sequence model and determine the normal time for the following work activity: A worker takes 2 steps, grasps the waist-level feed lever on the lathe, pulls up the lever approximately 15 cm to engage the feed. The process time to turn the part is 25 sec. There is no alignment and no action by the worker at the end of the process time.

Solution: Referring to Tables 6 and 7, and multiplying the 25 sec process time by 2.78 to obtain a value of 69.5 for the X parameter (rounded up to 70), we obtain the following indexed activity sequence model for this work activity:

$$A_3\ B_0\ G_1\ M_1\ X_{70}\ I_0\ A_0$$

where A_3 = take 2 steps, B_0 = no body motion, G_1 = gains control of lever, M_1 = pull the lever up 15 cm, X_{70} = process time of 25 sec, I_0 = no alignment, and A_0 = no motion. The sum of the index values is 75. Multiplying by 10, we have 750 TMUs (about 27 sec). The lathe process time accounts for all but 2 sec of the activity sequence. ■

Tool Use. The Tool Use sequence model applies to the variety of work situations that involve the use of a hand tool. Hand tools covered by this model include screwdrivers, ratchets, hammers, scissors, knives, various kinds of wrenches, and common measuring instruments. When a worker's hand is used to perform a motion such as turning a screw, Tool Use is employed. The sequence model also covers writing, marking, reading, thinking, and inspecting. The parameters in the Tool Use sequence include A, B, G, and

P that were previously used in the General Move and Controlled Move. Only one new parameter is included in Tool Use, and that refers to the specific action that applies to the tool being used in the activity sequence. The possible parameters include the following:

F—Fasten. This parameter is used to describe any fastening motion that is performed with the fingers, hand, or a tool such as a screwdriver or wrench.

L—Loosen. This parameter is similar to Fasten, except the objective is to unfasten or disassemble an object.

C—Cut. The Cut parameter describes the manual actions required to slice, slit, or otherwise separate an object. Common tools for cutting include knives, scissors, and pliers.

S—Surface treat. This parameter is used for applying a coating to the surface of an object (e.g., using a brush) or to remove a material from the surface (e.g., wiping the surface).

M—Measure. The use of measuring instruments (e.g., linear scales, calipers, micrometers) or gauges is modeled by this parameter.

R—Record. The manual actions involving the use of a pen, pencil, or other marking tool to record data are covered by this parameter.

T—Think. This parameter models the visual and mental activities required to read or inspect an object.

Only one of these parameters is included in the standard sequence for the Tool Use activity sequence model. Its placement in the sequence model is indicated by the asterisk (*) in the following:

A B G A B P * A B P A

where the first three parameters (A B G) are the motions to ***get*** the tool, the next three parameters (A B P) ***put*** or place the tool into position, * is the tool use code (F, L, C, S, M, R, or T), the next three parameters (A B P) ***put*** the tool aside, and the final parameter (A) is to provide for the worker to ***return*** to some previous position. Table 8 (a) and (b) list the index values for the Tool Use parameters.

Example 3 Tool Use

Develop the activity sequence model and determine the normal time for the following work activity: A worker picks up a screw from his worktable, positions it into a threaded hole, and turns it three spins with his fingers. He then picks up a screwdriver from the worktable, positions it on the head of the screw, fastens the screw with six turns, and lays the screwdriver aside.

Solution: This requires two activity sequence models, corresponding to the two sentences of the work description. The first model is developed as follows:

$$A_1\ B_0\ G_1\ A_1\ B_0\ P_3\ F_6\ A_0\ B_0\ P_0\ A_0$$

TABLE 8 (a) MOST Parameters and Index Values for Tool Use—Fasten and Loosen

Fasten or Loosen = **F** or **L**												
	Finger Action	Wrist Action				Arm Action					Tool Action	
Index	Fingers, Screw-driver	Hand, Screw-driver, Ratchet, T-wrench	Wrench, Allen key	Wrench, Allen key, Ratchet	Hand, Hammer	Ratchet	T-wrench, 2 hands	Wrench, Allen key	Wrench, Allen key, Ratchet	Hand, Hammer	Power wrench	Index
	Spins	Turns	Strokes	Cranks	Taps	Turns	Turns	Strokes	Cranks	Strikes	Diameter	
1	1	—	—	—	1	—	—	—	—	—	—	**1**
3	2	1	1	1	3	1	—	1	—	1	6 mm (1/4 in.)	**3**
6	3	3	2	3	6	2	1	—	1	3	25 mm (1 in.)	**6**
10	8	5	3	5	10	4	—	2	2	5		**10**
16	16	9	5	8	16	6	3	3	3	8		**16**
24	25	13	8	11	23	9	6	4	5	12		**24**
32	35	17	10	15	30	12	8	6	6	16		**32**
42	47	23	13	20	39	15	11	8	8	21		**42**
54	61	29	17	25	50	20	15	10	11	27		**54**

TABLE 8 (b) Parameters and Index Values for Tool Use—Cut (C), Surface treat (S), Measure (M), Record (R), and Think (T)

		Cut = **C**			Surface Treat = **S**			Measure = **M**	Record = **R**		Think = **T**				
	Cutoff	Secure	Cut	Slice	Air clean	Brush clean	Wipe	Measure	Write		Mark	Inspect	Read		
	Pliers		Scissors	Knife	Nozzle	Brush	Cloth	Measuring Tool	Pen/Pencil		Marker	Eyes, fingers	Eyes		
Index	Wire		Cuts	Slices	.1 m^2 (1 ft^2)	.1 m^2 (1 ft^2)	.1 m^2 (1 ft^2)		Digits	Words	Digits	Points	Digits, single words	Text	Index
1		Grip	1						1		Check	1	1	3	**1**
3	Soft		2	1			1/2		2		1 Scribe line	3	3	8	**3**
6	Medium	Twist Form loop	4		1 Spot cavity	1			4	1	2	5 Feel for heat	6	15	**6**
10	Hard		7	3			1	Profile gauge	6		3	9 Feel defect	12	24	**10**
16		Secure cotter pin	11	4	3	2	2	Fixed scale, Caliper	9	2	5	14		38	**16**
24			15	6	4	3		Feeler gauge	13	3	7	19		54	**24**
32			20	9	7	5	5	Steel tape, Depth micrometer	18	4	10	26		72	**32**
42			27	11	10	7	7	OD micrometer	23	5	13	34		94	**42**
54			33					ID micrometer	29	7	16	42		119	**54**

where $A_1\ B_0\ G_1$ = pick up screw from worktable, $A_1\ B_0\ P_3$ = position screw into threaded hole, F_6 = turn screw three spins with fingers, and $A_0\ B_0\ P_0\ A_0$ = no motions. The second activity sequence model is the following:

$$A_1\ B_0\ G_1\ A_1\ B_0\ P_3\ F_{10}\ A_1\ B_0\ P_1\ A_0$$

where $A_1\ B_0\ G_1$ = pick up screwdriver from worktable, $A_1\ B_0\ P_3$ = position screwdriver on screw head, F_{10} = turn screwdriver six turns, $A_1\ B_0\ P_1$ = lay screwdriver aside, and A_0 = no motion.

The sum of the index values is 12 for the first model and 18 for the second, a total of 30. Multiplying by 10, we have 300 TMUs (about 10.8 sec). ■

3.2 Additional Versions of MOST

In addition to Basic MOST, several additional versions of MOST are available for other application characteristics. These are described in Table 9. A description of Basic MOST is included for comparison purposes.

3.3 MOST for Windows

Keeping up with trends in work measurement, H. B. Maynard and Company has developed a computerized version of MOST, called MOST for Windows. Depending on the application, MOST for Windows allows the user to analyze a given work situation by any of the three MOST analysis techniques: (1) Basic MOST, (2) Maxi MOST, or (3) Mini

TABLE 9 Other Versions of MOST Work Measurement Packages

Basic MOST: A system designed for manual work that involves the movement of objects. It is intended for applications in which the work cycle is performed more than 150 times per week but less than 1500 times per week. A typical cycle time for Basic MOST is 30 sec to 3 min. As described in Section 3.1, Basic MOST uses three activity sequence models: (1) General Move, (2) Controlled Move, and (3) Tool Use. The analyst time required to apply Basic MOST is about 10 times the task cycle time.

Maxi MOST: A system developed for applications in which the work cycle is performed fewer than 150 times per week and there are variations in the cycle. Task times of several hours can be analyzed using this version. Examples of work activity that can be analyzed by Maxi MOST include setup tasks and heavy assembly. Maxi MOST consists of five activity sequence models: (1) part handling, (2) tool/equipment use, (3) machine handling, (4) transport with powered crane, and (5) transport with wheeled truck. The analyst time required to apply Maxi MOST is 3 to 5 times the task cycle time.

Mini MOST: A system developed for operations in which the work cycle is highly repetitive and performed more than 1500 times per week. For a 40-hour week, this translates to a cycle time of 1.6 min or less. However, cycle times analyzed by Mini MOST are usually much less than this. Mini MOST consists of two activity sequence models: (1) General Move and (2) Controlled Move. The analyst time required to apply Mini MOST is about 25 times the task cycle time.

Clerical MOST: A system similar to Basic MOST but designed for clerical work activity. In addition to the three basic activity sequence models in Basic MOST (General Move, Controlled Move, and Tool Use), there is an additional model for equipment use. The format of the model is the same as Tool Use, except that additional parameters are available for actions involving the operation of office equipment.

Source: Estimates of analyst time to apply the MOST versions are based on [9].

MOST (Table 9). Other alternatives are also available. The analyst can use action distances (the A parameter in Basic MOST) and body motions (the B parameter) that are based on data about the layout of the workplace, rather than determining and inputting the index values directly.

Among the other options, MOST for Windows includes two modules that can be called by the analyst:

- *Quick MOST.* This feature permits the user to select among multiple possible work methods that might be applied to the given application. The method description, activity sequence model, and associated normal time are immediately provided by Quick MOST, where editing can be performed by the analyst if needed. The selection list of work methods is based on data of industrial applications compiled by Maynard. The user can add additional work methods representing the company's unique applications if desired.
- *Direct MOST.* This MOST for Windows feature automatically generates a description of the work method based on data provided about the work situation. The module allows the analyst to override or edit the automatically generated work method description when appropriate. Parameter index values are automatically developed by Direct MOST when the analyst selects rules that have been developed for MOST applications. These rules are available to the user through drop-down menus, or by specification of workplace objects and locations.

The following are the benefits and advantages reported by Maynard for users of MOST for Windows: (1) A complete methods description is provided. (2) Nonproductive work elements are identified during the analysis so that they can be eliminated from the standard method. (3) MOST for Windows is a simple and easy-to-use work measurement technique. (4) Developing time standards by MOST for Windows is efficient and accurate. (5) Database search capabilities are provided.

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REVIEW QUESTIONS

1 Define *predetermined motion time system*.
2 What are the steps in applying a predetermined motion time system?
3 What is the difference between a first-level PMTS and a higher-level PMTS?
4 What is a motion aggregate in a higher-level PMTS?
5 Compare the advantages of a higher-level PMTS and a first-level PMTS.
6 What is the unit of time used in methods-time measurement?
7 What does the acronym MOST stand for?
8 What is the primary focus of MOST in terms of type of work activity?
9 What are the motion aggregates in MOST called?
10 What is the difference between General Move and Controlled Move in MOST?

PROBLEMS

MTM-1

1 A worker seated at a table performs a REACH. The sought-after object is jumbled with other objects in a tote pan, and the distance of the reach is 18 in. Determine the MTM-1 symbol and normal time in TMUs for this motion element.
2 An assembly worker standing at a workbench performs a MOVE. The object being moved weighs 10 lb. It is moved to an exact location a distance of 20 in. Determine the MTM-1 symbol and normal time in TMUs for this motion element.
3 An office worker sitting at a desk performs a MOVE. The moved object is a file weighing less than 1 lb. The distance is an approximate location 14 in. away from the starting point. Determine the MTM-1 symbol and normal time in TMUs for this motion element.
4 A mechanic performs two POSITION elements in sequence. The first requires an alignment along longitudinal axes of the objects, and the second requires an orientation of their rotational axes. Both elements can be classified as close fits with no symmetry, and the objects are easy to handle. (a) Determine the MTM-1 symbols and normal times in TMUs for these motion elements. (b) What is the total time for both elements in seconds?
5 What is the MTM-1 normal time in TMUs that should be allowed an operator who must read an instruction set before proceeding to perform a processing task, in which the instruction document contains 150 words? How many seconds is this?
6 In performing a certain manual task, a worker must walk a distance of 30 ft (one way) as one of the elements. (a) What is the MTM-1 normal time in TMUs that should be allowed for the element? (b) What is this walking time in seconds?
7 For safety reasons, a worker's eye travel time in a certain operation must be separated from the manual elements that follow. The distance the worker's eyes must travel is 20 in. The perpendicular distance from her eyes to the line of travel is 2 ft. No refocus is required. What is the MTM-1 normal time in TMUs that should be allowed for the eye travel element?

8 A work element in a manual assembly task consists of the following MTM-1 elements: (1) R16C, (2) G4A, (3) M10B5, (4) RL1, (5) R14B, (6) G1B, (7) M8C3, (8) P1NSE, and (9) RL1. (a) Determine the normal times in TMUs for these motion elements. (b) What is the total time for this work element in seconds?

9 A work element in a machine maintenance operation consists of the following MTM-1 elements: (1) W5P, i.e., walk 5 paces, (2) B, i.e., bend (3) R14B, (4) G1A, (5) AB, i.e., arise from bend, (6) W5P, (6) M12B2, (7) P1SSE, (8) RL1. (a) Determine the normal times in TMUs for these motion elements. (b) What is the total time for this work element in seconds?

MOST

10 Develop the activity sequence model and determine the normal time for the following work activity: A worker walks 3 steps, picks up a screwdriver from the floor, returns to his original position, and places the screwdriver on his worktable.

11 Develop the activity sequence model and determine the normal time for the following work activity: A clerk walks 8 steps, bends and picks up a file folder from the floor, places it on the counter within reach at that location, and then returns to her original location.

12 Develop the activity sequence model or models and determine the normal time for the following work activity: An assembly worker obtains 4 bolts in one hand from a bin located 10 in. away on his worktable and puts one bolt each into four holes in the bracket within easy reach in front of him.

13 Develop the activity sequence model and determine the normal time for the following work activity: A machinist standing in front of his milling machine grasps the waist level feed lever on the machine, and rotates the lever 1 crank to engage the feed. The process time to mill the part is 50 sec. There is no alignment and no action by the worker at the end of the process time.

14 Develop the activity sequence model and determine the normal time for the following work activity: A material-handling worker grasps a carton weighing 20 lb on a counter and slides it along the countertop a distance of 2 ft.

15 Develop the activity sequence model and determine the normal time for the following work activity: A drill press operator reaches 20 cm (8 in.) and pulls the feed lever down to engage the feed motion, which takes 12 sec.

16 Develop the activity sequence model and determine the normal time for the following work activity: A worker walks three steps, picks up a screw from his worktable, walks back to his initial location, positions the screw into a threaded hole, and turns it 5 spins with his fingers.

17 Develop the activity sequence model and determine the normal time for the following work activity: A worker picks up a screwdriver within reach from his worktable, positions it onto the head of a screw, fastens the screw with 6 turns, and lays the screwdriver aside.

18 Develop the activity sequence model and determine the normal time for the following work activity: An assembly worker on a production line obtains an Allen key within reach, positions it 15 cm (6 in.) onto a bolt head, cranks it 7 times to seat the bolt, and then sets the key aside.

19 Express the MTM-1 motion elements in Problem 8 as one or more MOST activity sequence models with index numbers. (a) Determine the normal times in TMUs for these activity sequence models. (b) What is the total time for this (these) activity sequence model(s) in seconds? (c) How do the MOST normal times compare with the normal times from MTM-1?

20 Express the MTM-1 motion elements in Problem 9 as one or more MOST activity sequence models with index numbers. (a) Determine the normal times in TMUs for these

activity sequence models. (b) What is the total time for this (these) activity sequence model(s) in sec? (c) How do the MOST normal times compare with the normal times from MTM-1?

21 A work element in a worker-machine cycle has been reduced to the following two MOST activity sequence models:

$$A_3 B_{10} G_3 A_6 B_{10} P_3 A_3$$

$$A_3 B_0 G_1 M_1 X_T I_0 A_3$$

In the second sequence model, the process time is known to be 9.3 sec. This value must be converted to the correct index value, symbolized by the subscript T for the X parameter in the second sequence model. (a) What is the correct value of T? (b) Determine the normal time in seconds that would be allocated for this work element.

Miscellaneous

22 The normal time for walking in MTM-1 is 5.3 TMU/ft distance. (a) Using this value, determine the amount of time in minutes that would be required to walk 1 mile (5280 ft). (b) How does this compare with the traditional benchmark of standard performance of walking at 3 miles/hr?

23 The normal time for walking in MOST is 6.9 TMU/ft distance.[6] (a) Using this value, determine the amount of time in minutes that would be required to walk 1 mile (5280 ft). (b) How does this compare with the traditional benchmark of standard performance of walking at 3 miles/hr?

[6]Based on tabulated values in [9, p. 35].